

RESEARCH ARTICLE OPEN ACCESS

Machine Learning Classification of AI-Generated and Human-Mapped Buildings in OpenStreetMap Using Morphometric Analysis

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Received: 8 January 2026 | **Revised:** 17 February 2026 | **Accepted:** 20 February 2026

Keywords: AI-generated buildings | building footprints | crowdsourced mapping | geometric analysis | machine learning classification | morphometric analysis | OpenStreetMap

ABSTRACT

The proliferation of AI-generated building footprints in OpenStreetMap (OSM) has transformed crowdsourced mapping, yet the geometric characteristics associated with different digitization methods remain poorly understood. This study presents a comprehensive morphometric analysis of more than 9 million building footprints across 15 geographically diverse cities spanning six continents. To investigate whether buildings labeled with AI source tags exhibit distinct geometric patterns, machine learning classifiers were trained on a regionally balanced dataset of approximately 1 million buildings. Thirty-two shape-based features encompassing size metrics, shape regularity, complexity measures, and anomaly detection indicators were extracted, and three gradient boosting classifiers were evaluated to distinguish these contributions. The best-performing model (LightGBM with class weighting) was used to achieve 74.5% balanced accuracy, 82.9% recall, and 0.819 AUC, providing evidence that buildings labeled with AI source tags exhibit geometric patterns that differ systematically from other buildings in the dataset. Feature importance analysis revealed that orthogonality measurements, vertex density patterns, and shape regularity indices were the strongest discriminators, with orthogonality z-scores alone accounting for 18.2% of model importance. AI-generated buildings showed significantly higher orthogonality (angles within 5° of 90°), greater rectangularity, and more consistent vertex spacing compared to human-digitized footprints. Geographic analysis revealed substantial variation in both AI adoption (0.15% in Berlin to 15.7% in Cairo) and model performance across regions, with balanced accuracy ranging from 67.5% (Asia) to 79.9% (Oceania). However, 34% of human-mapped buildings exhibited AI-like geometric patterns: an overlap reflecting multiple competing factors including methodological convergence, label uncertainty from hybrid workflows, OSM's version-history limitations, and natural variation in human digitization practices, none of which can be definitively separated without independent ground truth validation. These findings provide exploratory evidence that geometric analysis might complement other approaches to understanding data provenance in crowdsourced platforms, though substantial overlap between categories limits the utility of morphometric features for definitive source attribution.

1 | Introduction

OpenStreetMap (OSM), the world's largest crowdsourced geographic database, has undergone a fundamental transformation with the integration of artificial intelligence into its

mapping workflow. Since its inception in 2004, OSM has relied primarily on volunteer mappers who manually digitize geographic features from satellite imagery, GPS traces, and local knowledge (Haklay and Weber 2008). However, the emergence of AI-assisted mapping tools, particularly automated

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building footprint extraction systems developed by Microsoft, Google, Meta, and others, has dramatically accelerated the pace of map completion while simultaneously introducing new questions about data quality, geometric consistency, and the nature of crowdsourced geographic information (Herfort et al. 2023; Fila et al. 2025).

The proliferation of AI-generated building footprints in OSM represents both an opportunity and a challenge for the geospatial community. On one hand, machine learning models trained on satellite imagery can rapidly digitize millions of building structures, dramatically reducing the time and human effort required to create base maps in data-sparse regions (Anderson et al. 2019). On the other hand, the geometric characteristics of these automated contributions may differ systematically from human-digitized features in ways that affect data usability, cartographic quality, and downstream applications ranging from disaster response to urban planning (Dorn et al. 2015; Minghini and Frassinelli 2019). The ability to distinguish these contributions matters for two reasons: operationally, it enables quality control systems to prioritize validation resources toward buildings exhibiting geometric anomalies likely requiring human review; and strategically, it provides empirical evidence for developing governance frameworks as crowdsourced platforms integrate automation at scale. Understanding these geometric differences is essential for developing quality control systems, assessing the fitness-for-purpose of OSM data, and ensuring that the integration of AI tools enhances rather than compromises the integrity of crowdsourced geodata.

This intrinsic geometric approach offers distinct methodological advantages over comparative studies that rely on external reference datasets, as it requires only the geometric properties inherent to the building footprints themselves, making it applicable anywhere in OpenStreetMap without requiring access to authoritative cadastral data or ground truth validation datasets that are often unavailable in data-sparse regions. However, this approach cannot independently validate the accuracy of source tag labels and thus represents an exploratory rather than definitive analysis of digitization provenance. Buildings classified as AI-generated or human-mapped based on their OSM source tags reflect contributor-reported metadata rather than independently verified ground truth and may not fully capture hybrid workflows where AI outputs receive human editing, instances of missing or incomplete tags, or variations in tagging conventions across different mapping campaigns.

Despite the widespread adoption of AI-generated building footprints, the morphometric characteristics that distinguish automated from human-mapped contributions remain poorly understood. Previous research on OSM data quality has largely focused on positional accuracy, completeness, and semantic consistency (Barrington-Leigh and Millard-Ball 2017; Girres and Touya 2010). Recent studies have advanced methodologies for assessing OSM building completeness at scale, including population-based assessment approaches (Zhang et al. 2022), global city-scale completeness evaluations covering nearly 13,000 cities (Zhou et al. 2022), analyses of building data evolution and spatial patterns (Tian et al. 2019), and density-based quality estimation methods (Zhou 2018). However, while the

geometric signatures of different mapping methods have received limited attention. The few studies that have examined building footprint geometry have typically compared accuracy against authoritative datasets rather than investigating the intrinsic geometric patterns that emerge from different digitization approaches (Hecht et al. 2013; Brovelli and Zamboni 2018). This knowledge gap warrants investigation, particularly because geometric regularity directly affects applications such as 3D building modeling, change detection, and urban morphology analysis (Brasebin et al. 2012).

The present study addresses this gap through a comprehensive morphometric analysis of over 9 million building footprints across 15 geographically diverse cities spanning six continents. Drawing on this complete dataset, stratified regional sampling was employed to create a machine learning dataset of approximately 1 million buildings to test the hypothesis that distinctive geometric patterns reflecting algorithmic origin characterize AI-generated buildings compared to human-digitized structures. To test this hypothesis, 32 shape-based features were extracted based on theoretical principles regarding how algorithmic regularization, digitization granularity, and multidimensional optimization distinguish automated from manual mapping (detailed rationale in Section 3.3). Spanning size metrics, regularity measures, complexity indicators, and anomaly detection scores, these features were used to evaluate three gradient boosting classifiers (Random Forest, XGBoost, and LightGBM) in distinguishing AI from human contributions.

Three principal contributions are made to the understanding of automated mapping in crowdsourced geodatabases. First, the first large-scale, multicontinental quantification of the geometric differences between AI-generated and human-mapped building footprints is provided, revealing that these two digitization methods produce statistically distinct morphometric signatures. Second, the specific geometric features (particularly orthogonality measurements, vertex density patterns, and shape regularity indices) that most strongly discriminate between automated and manual mapping approaches are identified. Third, substantial geographic variation in both AI adoption rates and model performance is documented, demonstrating that the effectiveness of geometric classification depends critically on local mapping contexts and urban morphologies.

These findings have important implications for OSM data quality assessment and the future of crowdsourced cartography. The ability to detect patterns associated with AI source tags based solely on geometric properties offers a complementary approach to metadata-based quality control, enabling systems to flag potentially problematic footprints, identify mapping inconsistencies, and assess the heterogeneity of data sources within a geographic area. However, this exploratory approach reveals both the potential and limitations of morphometric analysis for understanding data provenance. Understanding whether and how geometric patterns can reliably distinguish different digitization methods provides insight into the broader question of how human mappers and algorithmic systems are reshaping crowdsourced geographic information, a question essential for developing governance frameworks and quality standards as AI becomes increasingly embedded in collaborative mapping workflows.

The remainder of this paper is organized as follows: The background and the related studies are presented in Section 2. Data acquisition, preprocessing pipeline, feature engineering approach, and machine learning methodology are described in Section 3; classification results, feature importance analysis, and geographic performance patterns are presented in Section 4; and the implications of the findings for OSM data quality, the geometric footprint of mapping automation, and the evolution of crowdsourced geodata in the age of artificial intelligence are discussed in Section 5; and conclusions synthesizing the key findings and directions for future research are presented in Section 6.

2 | Background

2.1 | OpenStreetMap and Volunteered Geographic Information

OpenStreetMap, launched in 2004, represents the largest and most successful crowdsourced geographic database globally, embodying the principles of Volunteered Geographic Information (VGI). The platform relies on a distributed network of volunteer mappers who manually digitize geographic features from satellite imagery, GPS traces, and local knowledge (Haklay and Weber 2008). This collaborative approach has proven remarkably effective, with OSM achieving over 80% completeness in road network mapping worldwide (Barrington-Leigh and Millard-Ball 2017). However, the quality and characteristics of VGI data have been subject to extensive scrutiny, as the diversity of contributors, mapping methodologies, and technological tools introduces significant heterogeneity into the resulting datasets.

Research on OSM data quality has traditionally focused on three primary dimensions: positional accuracy, completeness, and semantic consistency. Girres and Touya (2010) established foundational quality assessment frameworks comparing OSM data against authoritative reference datasets, revealing that positional accuracy varies substantially across geographic contexts and feature types. Hecht et al. (2013) investigated the spatio-temporal patterns of building footprint completeness in OSM, demonstrating that data maturity follows distinct evolutionary trajectories influenced by local mapping communities and organized mapping campaigns. These traditional quality assessment approaches typically require external reference datasets for validation, limiting their applicability to regions where authoritative data exists. More recently, Herfort et al. (2023) conducted a comprehensive spatiotemporal analysis of global urban building data in OSM, revealing persistent inequalities in data coverage that reflect broader patterns of digital divide and resource distribution.

Recent work has highlighted both the progress and challenges in OSM building data completeness. Ullah et al. (2023) developed crowdsourcing approaches using the MapSwipe mobile application to assess OSM building completeness, finding substantial heterogeneity across regions. Beyond geometric completeness, semantic enrichment of OSM building data has emerged as a critical research direction: Memduhoğlu et al. (2024) demonstrated that Large Language Model embeddings of OSM tags can significantly improve building function classification,

addressing the persistent challenge that while OSM provides excellent geometric coverage, descriptive attributes remain sparse. In conflict-affected areas, dedicated mapping campaigns have demonstrated that carefully coordinated human mapping can outperform AI-generated datasets. The humanitarian OpenStreetMap Team's 2024 Gaza mapping project produced 18% more buildings than Microsoft's AI-generated footprints, while the subsequent Lebanon initiative produced 65% more buildings in the affected districts (HOT 2024a; 2024b). These findings underscore ongoing tensions between automated extraction efficiency and human-verified quality in critical-use scenarios.

Despite this extensive body of research on OSM data quality, the geometric signatures that distinguish different digitization methodologies have received limited attention. Most quality assessments have compared OSM data against authoritative cadastral datasets (Brovelli and Zamboni 2018) or evaluated semantic accuracy and feature completeness (Dorn et al. 2015), while the intrinsic morphometric patterns emerging from human versus automated mapping practices remain poorly characterized. This knowledge gap has become increasingly critical as AI-generated contributions now constitute a substantial and growing fraction of OSM's building inventory.

2.2 | AI-Assisted Mapping and Automated Building Footprint Extraction

The integration of artificial intelligence into crowdsourced mapping workflows represents a fundamental paradigm shift in how geographic data are produced and maintained. Beginning in the mid-2010s, major technology companies including Microsoft, Google, Meta, and Maxar developed machine learning systems capable of automatically extracting building footprints from satellite and aerial imagery at unprecedented scales (Anderson et al. 2019). These AI systems, typically based on convolutional neural networks trained on millions of labeled building examples, can process entire countries in days, a task that would require thousands of volunteer hours through traditional manual digitization.

The proliferation of AI-generated building footprints has been substantial and continues to accelerate. A comprehensive 2025 analysis by Fila, Štampach, and Herfort revealed that as of January 2024, Microsoft Building Footprints alone accounted for 20.3 million buildings (3.45% of all OSM buildings), with Google Open Buildings contributing additional millions primarily across Africa and Asia. The study documented substantial geographic variation in AI adoption, ranging from minimal presence in traditional OSM strongholds like Western Europe to dominant market share in previously undermapped regions. However, this rapid integration raises important methodological concerns, including model bias across different geographic contexts, lack of transparency in proprietary algorithms, and absence of feedback loops for continuous model improvement (HOT 2024c).

In response to these challenges, the Humanitarian OpenStreetMap Team developed fAIr (Free AI for Resilience), an open-source AI-assisted mapping service that enables local

communities to train and deploy custom building extraction models tailored to their specific geographic and cultural contexts (HOT 2024d). This approach addresses three critical limitations of commercial AI building datasets: algorithmic transparency, geographic bias from training data, and integration of human feedback for model refinement.

The technical foundations of automated building extraction have advanced significantly through deep learning innovations. Deep learning architectures including U-Net and DeepLabv3 have demonstrated effectiveness for automated building extraction from remote sensing imagery (Ji et al. 2019; Wang et al. 2023). Dabove et al. (2024) demonstrated that data fusion approaches combining RGB orthophotos with digital surface models can further improve segmentation accuracy, while specialized post-processing methods address persistent challenges in boundary regularization and polygon vectorization (Girard and Tarabalka 2018).

These technical advances represent part of a broader paradigm shift toward what Song et al. (2023) and Sun et al. (2026) characterize as “AI-assisted mapping” ecosystems, where deep learning is embedded directly into mapping workflows to automate feature extraction and accelerate data updates. Recent end-to-end frameworks such as DeepMapper utilize Generative Adversarial Networks to detect changes and regularize building footprints, streamlining the transition from satellite imagery to vector data with minimal human intervention (Niroshan and Carswell 2025a). Concurrently, web-based applications are increasingly leveraging Large Language Models and SQL-based algorithms to facilitate interactive change detection and statistical retrieval, further lowering the technical barrier for contributors (Dsouza et al. 2025; Huang et al. 2025).

However, this rapid adoption of algorithmic map production introduces significant challenges regarding data transparency and trust. As AI-generated features become integrated into the OSM database, the distinction between human-verified contributions and machine-generated approximations becomes increasingly blurred (Andorful et al. 2025). This opacity has prompted calls for robust quality assurance frameworks capable of evaluating the morphometric fidelity and positional accuracy of AI-generated data (Niroshan and Carswell 2025b). The geometric characteristics of these AI-generated outputs, particularly their shape regularity, vertex patterns, and boundary complexity, may differ systematically from human-digitized features in ways that affect downstream applications ranging from 3D building modeling to urban morphology analysis (Brasebin et al. 2012). Understanding these geometric distinctions has thus become essential not only for quality control but also for maintaining the transparency and trustworthiness that underpin crowdsourced mapping initiatives.

2.3 | Morphometric Analysis of Building Footprints

The quantitative characterization of building geometry through morphometric analysis has a rich tradition in urban geography, cartography, and computational geometry. Shape descriptors provide mathematical tools for measuring and comparing the

geometric properties of spatial features, enabling objective assessment of form complexity, regularity, and similarity. In the context of building footprints, morphometric measures capture fundamental properties including size (area, perimeter), compactness (how closely a shape approximates a circle), rectangularity (deviation from rectangular form), and complexity (boundary irregularity and detail).

Polsby and Popper (1991) introduced the concept of compactness as a scale-independent measure of shape regularity, originally developed to detect gerrymandering in electoral districts but subsequently applied across numerous spatial analysis domains. Their compactness metric, defined as $4\pi A/P^2$ where A is area and P is perimeter, provides an intuitive measure ranging from 0 (highly irregular) to 1 (perfect circle). Basaraner and Cetinkaya (2017) systematically evaluated multiple shape indices for characterizing building footprints in GIS, identifying the Equivalent Rectangular Index (ERI) as particularly effective for quantifying how closely a building approximates its minimum bounding rectangle (a property that may prove diagnostic of algorithmic versus manual digitization).

Recent advances in 3D building modeling have expanded morphometric analysis beyond traditional 2D footprint measures. Labetski et al. (2023) introduced comprehensive 3D building metrics encompassing geometric properties (vertex counts, surface areas), derived properties (volume calculations, aspect ratios), and sophisticated space indices including circularity, convexity, fractality, and cuboidness. Their analysis of four Dutch cities revealed that detailed 3D geometry enables more nuanced characterization of architectural style and urban form than traditional height-only approaches. Similarly, Biljecki and Chow (2022) demonstrated that building morphometrics computed at scale enable quantitative comparisons of urban typologies across cities and continents, linking geometric patterns to phenomena including energy consumption, microclimate effects, and perceived urban vitality.

Contemporary research increasingly applies machine learning to morphometric data for urban analysis tasks. Wang et al. (2024) employed unsupervised visual representation learning on figure-ground maps enriched with morphology metrics to discover urban form typologies across San Francisco, Singapore, Amsterdam, and Barcelona, demonstrating that geometric features encode information about neighborhood character and development patterns. Durst et al. (2024) used building footprint morphometrics to examine relationships between neighborhood form and socioeconomic characteristics across five major US metropolitan areas, revealing systematic associations between geometric regularity and demographic patterns. These studies underscore morphometry's potential for revealing the social, economic, and technological processes that shape urban landscapes, including the growing influence of automated mapping systems.

Fractal dimension, introduced and formalized by Mandelbrot (1967, 1983) to quantify the complexity of irregular boundaries, offers another powerful tool for geometric analysis. Fractal measures capture how boundary detail changes with measurement scale, providing insight into the smoothing and simplification operations that automated systems often apply

during post-processing. Similarly, vertex metrics (including vertex count, vertex density, and vertex spacing patterns) provide direct evidence of digitization granularity and can reveal the systematic differences between human mappers who adapt vertex placement to local geometric detail and AI systems that may apply consistent vertex spacing or simplification algorithms.

More sophisticated shape descriptors capture multidimensional geometric properties. Elongation measures, developed by Dražić et al. (2010) and Stojmenović and Žunić (2007), quantify how stretched a shape is along its principal axis, while convexity ratios measure the degree of boundary concavity. Orthogonality metrics (quantifying the percentage of angles that approximate right angles) may prove particularly diagnostic of AI-generated buildings, as many automated extraction systems apply angle regularization to enforce perpendicular corners in building outlines. These morphometric tools, when combined systematically, enable comprehensive geometric fingerprinting of building footprints that can potentially distinguish automated from manual digitization approaches.

2.4 | Machine Learning for Geospatial Classification

The application of machine learning to geospatial classification tasks has become increasingly sophisticated, with gradient boosting methods emerging as particularly effective for handling complex, high-dimensional feature spaces characteristic of geographic data. Random Forests (Breiman 2001), XGBoost (Chen and Guestrin 2016), and LightGBM (Ke et al. 2017) represent state-of-the-art ensemble learning approaches that construct multiple decision trees and aggregate their predictions, offering robust performance across diverse classification problems. These methods excel at capturing nonlinear relationships between features and target classes, automatically handling feature interactions, and providing interpretable feature importance scores that reveal which geometric properties most strongly influence classification decisions.

A critical challenge in applying machine learning to OSM building classification is the severe class imbalance inherent in the dataset, where only 2.69% of the total building inventory in the study is composed of AI-generated buildings. Class imbalance can cause traditional classifiers to achieve high overall accuracy by simply predicting the majority class for all examples, while failing completely on minority class detection. Techniques for addressing class imbalance include undersampling the majority class, oversampling the minority class using methods like SMOTE, and applying class weights that penalize misclassification of minority examples more heavily than majority examples (Chawla et al. 2002). The choice of appropriate performance metrics is equally critical: balanced accuracy (the mean of per-class accuracies), F1-score (the harmonic mean of precision and recall), and the Matthews Correlation Coefficient provide more meaningful assessments than raw accuracy when classes are imbalanced.

Geographic variation in mapping practices and urban morphologies introduces additional complexity to classification tasks. Building characteristics may vary systematically across regions due to differences in architectural styles, urban planning

traditions, satellite imagery quality, AI tool deployment strategies, and local mapper conventions. Wu et al. (2024) showed that quantitative morphometric approaches enable robust comparison of building patterns across different cities when normalized appropriately. This geographic heterogeneity necessitates careful evaluation of model generalization: A classifier trained on data from diverse regions must demonstrate consistent performance across different geographic contexts to be useful for global-scale quality control applications. Hierarchical geographic analysis, examining performance patterns at both city and regional levels, can reveal whether classification difficulty varies systematically by location and identify contexts where automated detection may be more or less reliable.

3 | Materials and Methods

3.1 | Study Area and Data Acquisition

To evaluate the geometric distinctions between AI-generated and human-mapped building footprints, this study selected 15 geographically diverse cities across six continents: Europe, North America, South America, Africa, Asia, and Oceania (Table 1). City selection followed a two-step strategy: Cities were chosen to ensure regional representation across all six continents and to maintain a minimum of 1000 AI-generated buildings per city to ensure sufficient statistical power for reliable classification analysis. These locations were selected to represent a variety of urban morphologies and OSM tagging practices while balancing geographic diversity with data availability constraints. Raw OpenStreetMap data was acquired using the pyrosm Python library (version 0.6.1), which parses Protocolbuffer Binary Format (.osm.pbf) files. The dataset comprises a total of 9,116,074 building footprints distributed across 15 cities, with AI-generated buildings representing 2.69% of the total dataset (244,974 buildings). Regional variation in AI adoption was substantial, ranging from 0.15% in Berlin and Tokyo to 15.70% in Cairo.

3.2 | Data Preprocessing and Ground Truth Labeling

A rigorous preprocessing pipeline was implemented using GeoPandas (version 0.14.1) to ensure geometric accuracy and data quality. To minimize distortion in metric calculations (area, perimeter, angles), each city's dataset was reprojected from the global WGS84 coordinate system (EPSG:4326) to its optimal local Universal Transverse Mercator (UTM) zone using. Invalid geometries were then repaired using Shapely's function, while empty geometries and nonpolygon features were removed. MultiPolygon geometries were exploded into individual Polygon features to ensure consistent metric calculation across all buildings.

Buildings were classified into two categories (AI-generated and human-generated) based on the OSM source tag. A keyword-matching algorithm identified AI contributions using signatures including 'microsoft', 'buildingfootprints', 'google', 'globalml', 'ecopia', 'mapwithai', 'maxar', 'bing ai' and 'fair'. Buildings lacking a source tag or containing none of these keywords were classified as human-generated. This

TABLE 1 | Study cities with population, building counts, and AI adoption rates.

Region	City	Country	Population	AI buildings	Human buildings	AI percentage
Europe	Istanbul	Turkey	15.5 M	6743	592,133	1.13%
Europe	Berlin	Germany	3.7 M	1586	1,051,512	0.15%
Europe	London	UK	9.0 M	10,029	1,358,300	0.73%
Europe	Budapest	Hungary	1.7 M	34,104	445,168	7.12%
N. America	New York City	USA	8.3 M	47,024	1,540,373	2.96%
N. America	Toronto	Canada	2.8 M	12,436	711,102	1.72%
N. America	Dallas	USA	1.3 M	11,516	516,121	2.18%
N. America	Seattle	USA	0.8 M	74,432	481,702	13.38%
S. America	Rio de Janeiro	Brazil	6.7 M	2820	283,860	0.98%
S. America	Buenos Aires	Argentina	3.1 M	3058	146,358	2.05%
Africa	Cairo	Egypt	9.6 M	9143	49,098	15.70%
Africa	Cape Town	South Africa	4.6 M	5738	223,740	2.50%
Asia	Bangkok	Thailand	10.5 M	3186	237,612	1.32%
Asia	Tokyo	Japan	13.9 M	1556	1,017,621	0.15%
Oceania	Sydney	Australia	5.3 M	21,603	216,400	9.08%

Note: The 15 cities span six continents, with AI-generated buildings ranging from 0.15% (Berlin, Tokyo) to 15.70% (Cairo) of total building inventory.

classification approach has been validated by Fila et al. (2025) and aligns with documented AI mapping campaigns. This classification treats OSM source tags as ground truth, introducing inherent uncertainty. Source tags may be incomplete (undocumented AI tool usage), incorrect (human-edited AI buildings retaining AI tags), or ambiguous (hybrid workflows). If mislabeling is random, it reduces apparent accuracy; if systematic, it biases the model toward learning editing patterns rather than original digitization method. Without independent validation, this represents an exploratory study of tag-associated patterns rather than definitive provenance classification.

Following geometric validation, area-based filtering was applied to reduce noise and exclude nonbuilding artifacts. Geometries with an area smaller than 10 m² were removed after UTM projection, ensuring that the threshold was applied in meaningful metric units rather than decimal degrees. This filtering step removed spurious small features while retaining all legitimate building footprints.

As a final quality control step, buildings were flagged for quality issues including invalid geometry, self-intersections, extreme area values (> 1,000,000 m²), or excessive vertex counts (> 1000 vertices). Buildings failing these quality checks were excluded from analysis, resulting in a final dataset of 9,114,226 buildings representing a 99.98% retention rate.

3.3 | Feature Engineering: Shape Metrics

A high-dimensional feature vector was computed for every building footprint to capture morphological differences. A total

of 32 geometric indicators were selected for the machine learning analysis from an initial set of candidate features, retaining only those with ≥80% completeness across the dataset to ensure data quality, categorized into four morphometric groups (Table 2).

3.3.1 | Feature Selection Rationale

The selection of these 32 features was guided by three theoretical principles regarding how automated extraction systems should differ geometrically from human digitization. First, algorithmic regularization: AI systems typically apply post-processing operations to enforce geometric constraints, particularly right-angle snapping during polygon vectorization. Second, consistent digitization granularity: Automated systems apply uniform vertex placement algorithms whereas human mappers adapt vertex density to local architectural complexity and imagery quality. Third, multidimensional optimization: machine learning models trained on labeled building datasets may converge toward specific regions of morphometric feature space that optimize classification accuracy during training.

These principles informed selection across the four feature categories detailed below. Orthogonality metrics and their interactions directly test whether angle regularization creates detectable patterns. Vertex-based measures operationalize the granularity hypothesis by quantifying vertex spacing, density, and efficiency. Complexity indicators (fractal dimensions, boundary irregularity) assess whether systematic smoothing distinguishes the two categories. Derived features (z-scores, interaction terms, composite anomaly scores) test whether AI buildings occupy distinctive positions in multi-dimensional

TABLE 2 | Complete list of 32 geometric features organized by morphometric category.

Category	Count	Features
3.3.1—Size and Dimensions	7	Area, Perimeter, Log(Area), Sqrt(Area), Perimeter-Area Ratio, Perimeter per Sqrt(Area), Area-Perimeter Ratio ²
3.3.2—Shape Regularity	10	Rectangularity, Rectangularity (Manual), Squareness, Orthogonality, Circular Compactness, Polsby–Popper Compactness, Compactness Ratio, Shape Index, Convexity, ERI
3.3.3—Complexity & Angularity	9	Fractal Dimension (Momepy), Fractal Dimension (Standard), Vertex Count, Vertex Density, Corners, Vertex Efficiency, Elongation, Elongation ² , Orientation
3.3.4—Derived & Anomaly	6	Rectangularity–Orthogonality Interaction, Compactness–Fractal Interaction, Orthogonality Z-score, Rectangularity Z-score, Vertex Density Z-score, Anomaly Score
Total	32	

morphometric space that are invisible when examining features individually.

3.3.2 | Size and Dimensions (7 Features)

To account for scale-dependent variations and distribution skews, the basic Area and Perimeter were calculated, along with their normalized transformations: Log(Area) and Sqrt(Area). The Perimeter–Area Ratio and the Perimeter-per-Sqrt(Area) were also included to measure boundary scaling independent of absolute size.

3.3.3 | Shape Regularity and Compactness (10 Features)

This category captures how closely a building resembles standard geometric primitives. Rectangularity was computed as the ratio of the building's area to its Minimum Bounding Rectangle (MBR). This was calculated using two methods: a standard library implementation and a manual verification. Squareness quantifies the deviation from a perfect square. Orthogonality, a critical metric, measures the percentage of angles within 5° of 90°, which is particularly diagnostic of AI-generated buildings. Deep learning-based extraction systems commonly apply geometric regularization techniques including angle snapping and boundary orthogonalization during vectorization (Girard and Tarabalka 2018; Wang et al. 2023), potentially producing buildings with higher orthogonality than manual digitization.

For compactness measures, Polsby–Popper Compactness (Polsby and Popper 1991) was employed, which provides a scale-independent assessment ranging from 0 (highly irregular) to 1 (perfect circle). The Circular Compactness and Shape Index were also calculated. A Compactness Ratio (Polsby-Popper/Circular) was derived to detect specific shape distortions.

Convexity was measured as the ratio of the building area to its convex hull area. This metric quantifies the degree of concavity in a shape, with values approaching 1 indicating more convex (regular) forms. The Equivalent Rectangular Index (ERI)

was calculated following Basaraner and Cetinkaya (2017), who proposed this measure as one of the most effective shape complexity indicators for building footprints. ERI quantifies the similarity between a shape and its minimum rotated bounding rectangle, with values closer to 1 indicating more rectangular forms.

3.3.4 | Complexity and Angularity (9 Features)

To assess boundary irregularities and digitization granularity, Fractal Dimension was calculated using two distinct algorithms. The concept of fractal dimension, introduced and formalized by Mandelbrot (1967, 1983), quantifies the complexity of irregular boundaries by measuring how detail changes with measurement scale. Both the Momepy library method and a standard log–log approach based on the relationship $\log(\text{Perimeter})/\log(\text{Area})$ were implemented (Fleischmann 2019). Automated extraction systems may produce buildings with reduced boundary complexity compared to manual digitization, as post-processing algorithms apply smoothing operations to optimize polygon representation (Girard and Tarabalka 2018).

Vertex Metrics provide crucial discriminatory power between AI and human digitization practices. Total Vertex Count, Vertex Density (vertices per meter of perimeter), Vertex Efficiency (area per squared vertex count), and the number of Corners were calculated. Automated building extraction systems often apply vertex simplification algorithms during post-processing (Girard and Tarabalka 2018; Niroshan and Carswell 2025a), which may produce more consistent vertex spacing patterns compared to human digitization, where vertex placement varies with local geometric detail and mapper judgment.

Elongation measurements capture the aspect ratio and alignment of footprints. Elongation, calculated as the ratio of the longer to shorter edge of the minimum bounding rectangle (Dražić et al. 2010; Stojmenović and Žunić 2007), quantifies how stretched a shape is along its principal axis. Elongation Squared and Orientation were also calculated to capture various aspects of shape anisotropy.

3.3.5 | Derived Interactions and Anomaly Scores (6 Features)

To capture nonlinear “AI signatures” and outliers, several composite features were engineered. Interaction Terms amplify distinctive patterns: Rectangularity–Orthogonality Interaction emphasizes boxed shapes, Compactness–Fractal Interaction balances smoothness versus complexity, and Area–Perimeter Ratio Squared detects scale-dependent patterns.

To standardize critical metrics across different urban scales, Z-scores for Orthogonality, Rectangularity, and Vertex Density were calculated. These normalized measures allow direct comparison of building characteristics across cities with vastly different architectural styles and mapping densities. Finally, an Anomaly Score was computed as a composite distance metric summing the Z-scores of key indicators to quantify how far a building's geometry deviates from the statistical norm of the dataset.

3.4 | Machine Learning Classification Framework

3.4.1 | Sampling Strategy

Although morphometric features were extracted from all buildings in the preprocessed dataset, machine learning classification required careful handling of severe class imbalance to ensure reliable model training. To address severe class imbalance (AI buildings representing only 2.69% of the dataset) while preserving regional diversity, a stratified regional sampling approach that differed fundamentally from simple random undersampling was implemented. For each city, all AI-generated buildings were included to maximize signal from the minority class and preserve rare regional patterns that might be lost through random sampling. Human-digitized buildings were then sampled proportionally within each city to achieve an approximate 75:25 (human:AI) ratio. The 75:25 human-to-AI ratio deliberately introduces controlled imbalance to provide sufficient minority class representation for reliable model training while avoiding the unrealistic 50:50 balance that would poorly reflect deployment conditions. This ratio balances two competing needs: ensuring adequate AI building samples for the model to learn discriminative patterns (requiring higher AI prevalence than the 2.69% in the full dataset) while maintaining enough imbalance to prevent the model from learning decision boundaries optimized only for balanced data that would fail under realistic prevalence. This strategy ensured that regional geometric variations were captured across all 15 cities, that no rare AI-generated examples were excluded, that the training set remained computationally tractable (979,872 buildings total), and that geographic diversity was maintained in both training and test sets. The sampling was performed with a fixed random seed (seed = 42) to ensure reproducibility. The choice of seed value is arbitrary; any fixed seed would provide equivalent reproducibility while different seeds would yield statistically similar samples due to the large dataset size and stratified regional sampling design. This approach maintains the spatial and regional characteristics of the data, a critical consideration given that mapping practices and urban morphologies vary substantially across geographic contexts.

3.4.2 | Feature Preprocessing

Prior to model training, all 32 features underwent standardization using z-score normalization (mean=0, standard deviation=1) via scikit-learn's StandardScaler to ensure equal weighting across different measurement scales. The scaler was fit exclusively on the training set and then applied to the test set to prevent data leakage. Missing values, which remained in less than 20% of cases for retained features, were imputed using median values. Infinite values resulting from division operations were also replaced with column medians to ensure numerical stability during model training.

3.4.3 | Model Development

Three gradient boosting and ensemble learning approaches, each employing different strategies for handling class imbalance, were evaluated. The first approach combined a Random Forest classifier with random undersampling of the majority class using the imbalanced-learn library. This model utilized 100 trees with balanced class weights, a maximum depth of 12, square root feature sampling, and minimum sample requirements of 20 for splits and 10 for leaf nodes. The undersampling was set to a 50% sampling ratio, effectively reducing the majority class while maintaining all minority class samples through the pipeline.

The second approach employed XGBoost with class weighting, utilizing the parameter set to the inverse class ratio to automatically adjust loss functions for imbalanced data. This model used 150 trees with a maximum depth of 6 and a conservative learning rate of 0.05. To improve generalization and prevent overfitting, subsampling at 80% for both rows and columns was applied. The histogram-based tree construction method was employed for computational efficiency, with logloss specified as the evaluation metric.

The third approach utilized LightGBM with balanced class weighting, which automatically adjusts for class imbalance through its native class_weight parameter. This model also employed 150 trees with a slightly deeper maximum depth of 8, a learning rate of 0.05, and 31 maximum leaves per tree. Like XGBoost, it applied 80% subsampling for both rows and columns to improve generalization. All three models were trained using stratified 80/20 train-test splits (n_train = 783,897; n_test = 195,975) to maintain class proportions and regional representation across splits. Hyperparameters were selected to balance model complexity against overfitting risk, given the high-dimensional feature space of 32 features, with a fixed random seed ensuring reproducibility.

Hyperparameters were selected through iterative manual tuning prioritizing recall for the minority class while preventing severe overfitting. Learning rates (0.05) and tree depths (6–12) were constrained to balance model complexity against the risk of memorizing training patterns rather than learning generalizable geometric signatures. The different number of trees (100 for Random Forest vs. 150 for gradient boosting methods) reflects computational efficiency considerations, with Random Forest requiring fewer trees to stabilize predictions.

No formal grid search was conducted given computational constraints of the 9+ million building dataset; instead, parameters were selected based on established practices for imbalanced classification tasks in similar domains. The comparison of Random Forest with undersampling versus LightGBM and XGBoost with class weighting enables assessment of whether these different balancing strategies produce comparable discrimination ability or whether one approach is fundamentally superior for this task.

3.4.4 | Performance Evaluation

Model performance was assessed using a comprehensive suite of metrics appropriate for imbalanced binary classification. Balanced accuracy, calculated as the arithmetic mean of sensitivity (recall for the AI class) and specificity (recall for the human class), provided equal weight to both classes regardless of their prevalence in the dataset. Precision measured the proportion of predicted AI buildings that were truly AI-generated, while recall quantified the proportion of true AI buildings correctly identified by the model. The F1-score, computed as the harmonic mean of precision and recall, served as the primary metric for model selection as it balances both false positives and false negatives in a single measure.

The Matthews Correlation Coefficient (MCC) provides a balanced measure that accounts for all four cells of the confusion matrix and is particularly robust to class imbalance. The Area Under the ROC Curve (AUC) quantifies the model's discrimination ability across all classification thresholds, measuring how well the model separates the two classes. While overall accuracy (the proportion of correct predictions) was reported for completeness, it was not emphasized in model comparison due to the severe class imbalance that can make accuracy misleading.

Confusion matrices were computed and normalized by row (true labels) to examine class-specific error patterns, revealing whether the model struggled more with false positives (human buildings misclassified as AI) or false negatives (AI buildings misclassified as human). All metrics were calculated using the held-out test set, which represented 20% of the balanced sample and maintained the geographic and class distribution of the full dataset.

3.4.5 | Feature Importance Analysis

Feature importance scores were extracted from the trained Random Forest and LightGBM models using their native attributes, which measure each feature's contribution to reducing impurity (Gini impurity for Random Forest, gain for LightGBM) across all trees in the ensemble. For meaningful comparison across models with different scales, importance scores were normalized to sum to 100% within each model, representing each feature's percentage contribution to the total model importance.

To facilitate interpretation and identify broader patterns beyond individual features, the 32 features were grouped into four morphometric categories based on their geometric meaning

(as defined in Section 3.3). Size and Dimensions features captured area, perimeter, and their transformations. Shape Regularity and Compactness features included rectangularity, orthogonality, squareness, convexity, Polsby–Popper compactness, circular compactness, shape index, and the Equivalent Rectangular Index (ERI). Complexity and Angularity features encompassed fractal dimensions, vertex metrics including vertex count, vertex density, and vertex efficiency, corner detection, and elongation measures capturing aspect ratios and orientation. Derived Interactions and Anomaly Scores features included interaction terms (rectangularity–orthogonality, compactness–fractal), z-score normalized versions of key geometric properties (orthogonality, rectangularity, vertex density), and a composite anomaly score. Category-level importance was calculated by summing the normalized importance scores of constituent features within each category, revealing which geometric properties were most discriminative overall. The top 15 most important individual features were identified and analyzed in detail to understand which specific geometric characteristics most strongly distinguished AI from human contributions.

3.4.6 | Geographic Performance Analysis

To evaluate model generalization across different mapping contexts and assess whether classification performance varies systematically by location, hierarchical geographic performance analyses were conducted at both city and regional levels. For city-level analysis, all performance metrics (accuracy, balanced accuracy, precision, recall, F1-score, and AUC) were calculated using only test samples from each individual city. This approach revealed whether the model performed consistently across diverse urban contexts or showed systematic variation related to local mapping practices, urban morphology, or AI tool deployment patterns. Cities with fewer than 10 test samples or lacking representation of both classes were excluded from city-specific analysis to ensure statistical reliability and enable meaningful AUC calculation, which requires both classes to be present.

Regional performance metrics were computed by aggregating test predictions from all cities within each geographic region (Europe, North America, South America, Africa, Asia, Oceania). This hierarchical approach enabled examination of both fine-grained city-level patterns and broader regional trends in model discriminative ability. Such geographic variation could arise from several factors: region-specific digitization practices and mapper conventions, differences in AI tool deployment and parameterization across mapping campaigns, varying levels of mapper expertise and experience in different communities, or fundamental differences in urban morphology and building typologies that make some contexts inherently easier or harder to classify.

Performance patterns were visualized using multiple complementary approaches. Heatmaps displayed all metrics across cities simultaneously, enabling identification of cities where the model excelled or struggled. Bar charts facilitated direct comparison of regional performance, revealing geographic patterns in classification difficulty. Combined visualizations showed city-level F1-scores with cities color-coded by region, illustrating both individual city performance and regional clustering

patterns. These visualizations enabled identification of systematic geographic biases and helped assess the model's readiness for deployment across diverse global contexts.

3.5 | Statistical Comparison of Geometric Features

To formally test whether AI-generated and human-mapped buildings exhibit significantly different geometric characteristics, Mann–Whitney U tests were applied to compare the distribution of each of the 32 morphometric features between the two groups. The Mann–Whitney U test is a nonparametric alternative to the independent samples t -test and does not assume normal distribution of data, making it appropriate for geometric features that may exhibit skewed distributions.

Given the large sample size of buildings, statistical significance alone provides limited insight, as even trivially small differences may achieve $p < 0.05$. Effect sizes are therefore emphasized as indicators of practical importance. Effect sizes were calculated using the rank-biserial correlation coefficient (r), which ranges from -1 to 1 , where values of 0.1 , 0.3 , and 0.5 represent small, medium, and large effects respectively. Bonferroni correction was applied to adjust for multiple comparisons, a statistical safeguard that prevents false positives from accumulating when testing many hypotheses simultaneously (32 features in this case), by requiring a much stricter significance threshold ($\alpha = 0.05/32 = 0.00156$) to control the family-wise error rate across all feature tests.

Statistical analyses were performed separately for each geographic region to assess whether geometric differences between AI and human buildings varied across different urban morphologies and mapping contexts. All statistical tests were conducted using SciPy (version 1.11.3) in Python.

3.6 | Software and Computing Environment

All analyses were performed in Python 3.10 using a suite of specialized geospatial and machine learning libraries. GeoPandas (version 0.14.1) handled spatial operations including coordinate transformations and geometric manipulations. Momepy (version 0.7.2) provided specialized morphometric analysis functions for urban form. Scikit-learn (version 1.3.2) supplied machine learning pipelines, preprocessing tools, and evaluation metrics. The gradient boosting implementations came from XGBoost (version 2.0.3) and LightGBM (version 4.1.0), while imbalanced-learn (version 0.11.0) provided tools for handling class imbalance. NumPy (version 1.24.3) performed numerical operations, Pandas (version 2.0.3) enabled efficient data manipulation, and Matplotlib (version 3.7.2) and Seaborn (version 0.12.2) generated publication-quality visualizations.

Data processing scripts were implemented as modular, resumable pipelines with checkpoint saving to enable recovery from interruptions during long-running analyses of the 9+ million building dataset. All random operations, including train-test splits and undersampling, used fixed seeds ($\text{seed} = 42$) to ensure full reproducibility of results. This design allowed the entire analysis pipeline to be re-executed with identical results, a

critical requirement for scientific reproducibility. The code is provided as detailed in the Data Availability Statement.

3.7 | Methodological Limitations

The classification of buildings relies entirely on OSM source tags, which represent contributor-reported provenance rather than independently verified ground truth. This approach introduces several important limitations that affect the interpretation of our results. Buildings may be mislabeled due to incomplete tagging practices, where contributors fail to document data sources or apply tags inconsistently across different mapping campaigns. The binary classification into “AI-generated” and “human-mapped” categories does not capture hybrid workflows that are increasingly common in OSM, where AI-generated building footprints undergo subsequent human editing to correct errors, add missing details, or align boundaries with local knowledge. A building initially created by an AI system may be incorrectly labeled as human-contributed if the most recent editor did not preserve source metadata. Conversely, minor human edits to AI outputs may go undocumented, leaving buildings categorized as AI-generated despite subsequent manual refinement.

The dataset aggregates buildings created over multiple years, treating all AI contributions as equivalent despite substantial evolution in extraction algorithms, model architectures, and post-processing techniques during this period. Early automated extraction systems likely produced different geometric signatures than more recent models trained on larger datasets with improved regularization methods. Similarly, human mapping practices may have evolved in response to AI tool availability, with mappers potentially adopting more systematic digitization approaches over time. This temporal aggregation obscures longitudinal patterns and assumes static geometric relationships that may not hold across the dataset's temporal span.

The binary classification scheme groups outputs from multiple distinct AI systems (including Microsoft Building Footprints, Google Open Buildings, Meta/Maxar tools, and fAIr) into a single “AI-generated” category. These systems employ different neural network architectures, training datasets, and post-processing pipelines that likely produce varying geometric signatures. Microsoft's system, for example, may apply different angle regularization than Google's Open Buildings, while fAIr's community-trained models may exhibit region-specific characteristics absent in commercial tools. The observed geometric patterns therefore represent averaged signatures across heterogeneous tools rather than tool-specific characteristics, potentially masking important within-category variation that could inform more nuanced quality assessment approaches.

Model training is optimized by the sampling strategy to provide sufficient representation of the minority class, but the resulting 75:25 ratio does not reflect the actual distribution encountered in practical deployment scenarios. Model performance is evaluated on a regionally balanced test set with 25% AI prevalence, substantially higher than the 2.69% encountered in the full dataset. Class imbalance inherent in the dataset means that any practical deployment would face severe constraints on positive predictive value, limiting utility for definitive automated classification.

Hyperparameter selection was conducted through manual tuning rather than systematic grid search or cross-validation optimization. While the chosen parameters achieve reasonable classification performance, formal sensitivity analysis was not conducted to quantify how results vary with different parameter configurations. Geographic performance variation (Section 4.6) likely reflects inherent differences in data characteristics across regions, including local architectural styles, mapping conventions, and AI tool deployment patterns, rather than suboptimal parameter choices, since identical model architectures were applied uniformly across all cities. Future work should investigate whether region-specific hyperparameter tuning could improve classification accuracy in challenging contexts like Tokyo and Berlin.

4 | Results

4.1 | Model Performance and Classification Accuracy

Three gradient boosting classifiers were evaluated using 32 morphometric features to distinguish AI-generated from human-mapped buildings: Random Forest with undersampling, XGBoost with class weighting, and LightGBM with class weighting. Model performance was assessed using a regionally balanced test set comprising 195,975 buildings (48,994 AI-generated and 146,981 human-mapped), maintaining an approximate 25:75 AI-to-human ratio (Table 3).

It should be noted that these performance metrics reflect evaluation on a balanced test set where AI buildings constitute approximately 25% of the sample. These metrics would not transfer directly to real-world deployment, where the 2.69% prevalence of

AI buildings in the full dataset would substantially affect positive predictive value.

The LightGBM model with class weighting emerged as the best-performing classifier, achieving 74.46% balanced accuracy, 82.95% recall for AI buildings, 44.83% precision, and 0.819 AUC. The Random Forest model with undersampling achieved comparable performance (73.68% balanced accuracy, 80.68% AUC). In contrast, the XGBoost model achieved only 0.68% recall for AI buildings despite 93.24% precision, essentially classifying nearly all buildings as human-generated. This degenerate solution arises because XGBoost's `scale_pos_weight` parameter adjusts the loss function but does not prevent the optimizer from converging to the majority-class prediction when gradient signals from the minority class remain insufficient. Figure 1 visualizes these behaviors: LightGBM and Random Forest produce partially separated distributions whose substantial overlap, where many human buildings receive high AI-class probabilities, explains the precision ceiling. In contrast, XGBoost exhibits a degenerate pattern in which both classes are assigned near-zero probabilities; this confirms that its near-zero recall reflects a failure to learn discriminative patterns rather than a conservative threshold choice.

The three classifiers exhibited distinct strategic approaches to balancing precision and recall (Figure 2). The radar chart comparison reveals that the LightGBM and Random Forest models achieved similar overall performance profiles, with both prioritizing recall over precision, while XGBoost's distinctly different profile (high precision but near-zero recall) reflects its failure to detect the minority class. This recall-focused strategy in the first two models reflects the class imbalance challenge: maximizing detection of the minority AI class while accepting some false positives.

TABLE 3 | Overall model performance metrics on test set ($n = 195,975$ buildings).

Model	Balanced Acc.	Precision	Recall	F1-score	AUC	MCC	Accuracy
LightGBM	74.46	44.83	82.95	58.20	81.85	42.49	70.22
Random Forest	73.68	43.53	83.42	57.21	80.68	41.04	68.80
XGBoost	50.33	93.24	0.68	1.34	81.80	6.71	75.16

Note: Best scores in bold. Balanced Accuracy is the mean of per-class recall.

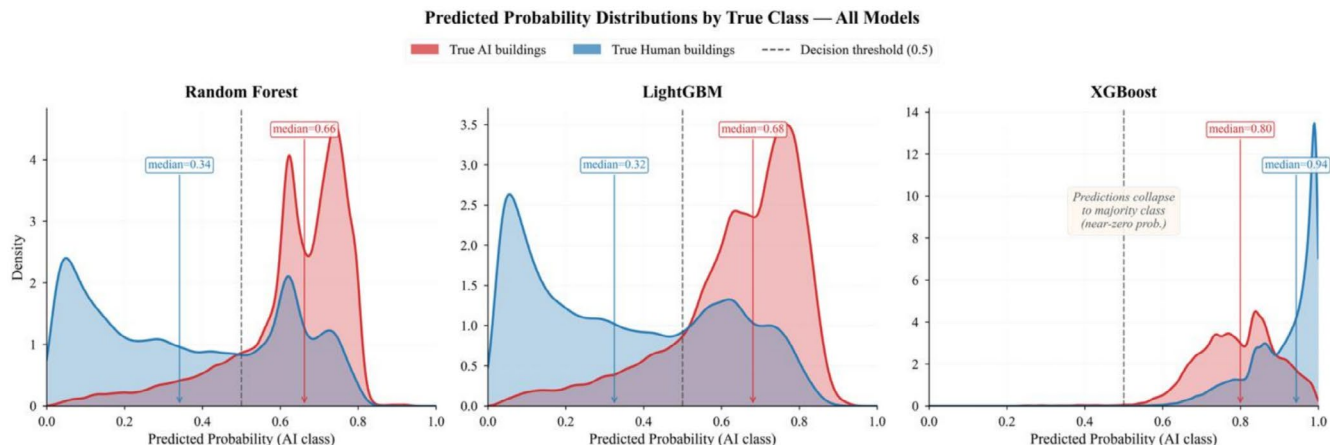


FIGURE 1 | Predicted probability distributions for the AI class across all three models, separated by true label. Red curves represent true AI-generated buildings; blue curves represent true human-mapped buildings.

The confusion matrices reveal the classification behavior of each model (Figure 3). The LightGBM model correctly classified 40,639 out of 48,994 AI buildings (82.95% recall) but misclassified 50,008 human buildings as AI-generated, resulting in 90,647 total positive predictions. The Random Forest model showed similar patterns with slightly higher recall (83.42%) but lower precision (43.53%). The XGBoost model's confusion matrix confirms its failure to learn discriminative patterns for the minority class.

4.2 | Classification Uncertainty and Label Ambiguity

The model misclassified 50,008 of 146,981 human buildings (34.0%) as AI-generated: a figure exceeding the total AI buildings in the test set (48,994). Conversely, 8355 of 48,994 AI-generated

buildings (17.1%) were classified as human-mapped. This substantial overlap admits multiple interpretations that cannot be disentangled without external validation: genuine geometric convergence where human mappers adopt regularization practices, source tag errors where mislabeled buildings cause apparent misclassification despite correct geometric pattern recognition, or hybrid workflows inadequately captured by binary labels. These patterns reveal that “AI-generated” and “human-mapped” represent overlapping distributions along a geometric regularity continuum rather than discrete, separable classes. For example, a human-digitized rectangular building with orthogonality=1.0, vertex density=0.095 vertices/m, and rectangularity=0.98 produces a morphometric signature geometrically indistinguishable from an AI-generated building with identical measurements; the model correctly identifies the geometric regularity pattern, but that pattern alone cannot reveal its origin. Figure 4 illustrates this overlap: The false positive example (bottom right) is geometrically indistinguishable from the AI-generated buildings above, sharing identical orthogonality (100%), comparable vertex density (0.113 v/m), and a similar fractal dimension (0.790). This demonstrates that geometric regularity is not exclusive to algorithmic extraction, rendering such false positives a structural inevitability rather than a model deficiency.

This overlap admits multiple interpretations that cannot be disentangled without external validation. It may reflect genuine methodological convergence, where human mappers increasingly adopt geometric regularization through automated tools or learned efficiency practices. Alternatively, it may expose label uncertainty, where source tags imperfectly capture hybrid workflows combining AI extraction with human editing, or where OSM's version history tracks only the most recent editor rather than original provenance. The variability within AI-generated buildings (with different systems producing distinct geometric signatures) further complicates classification. These patterns collectively demonstrate fundamental limitations in using geometric features alone for definitive provenance determination.

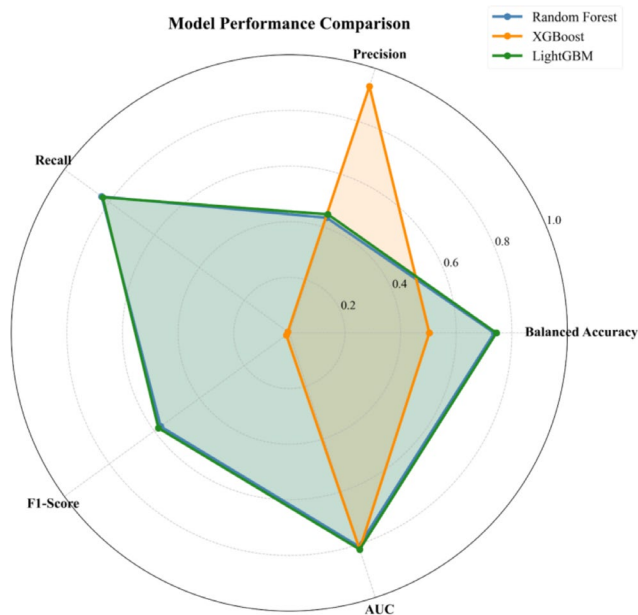


FIGURE 2 | Model performance profiles reveal distinct strategic approaches: LightGBM and Random Forest prioritize high recall (sensitivity to AI buildings) at the cost of precision, while XGBoost exhibits an inverted pattern. Each axis scaled 0–1, with larger polygons indicating better performance.

4.3 | Statistical Comparison of Geometric Properties

Mann-Whitney U tests revealed statistically significant differences ($p < 0.001$, Bonferroni-corrected $\alpha = 0.00156$) between

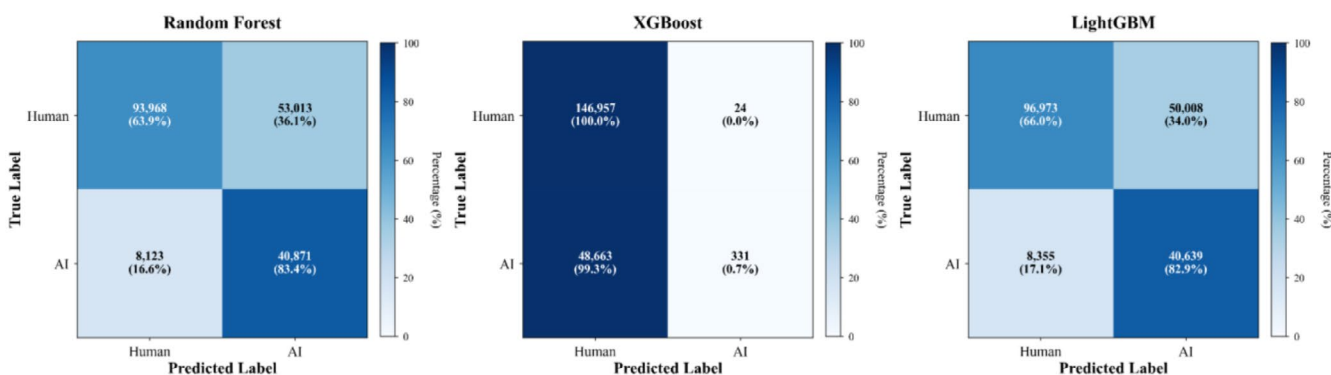


FIGURE 3 | Confusion matrices for Random Forest, XGBoost, and LightGBM models showing classification performance on test set ($n = 195,975$ buildings). Values show counts with row-normalized percentages in parentheses. These matrices reflect performance on a balanced test set (25% AI prevalence); under realistic prevalence (2.69% AI), positive predictive value would be substantially lower.

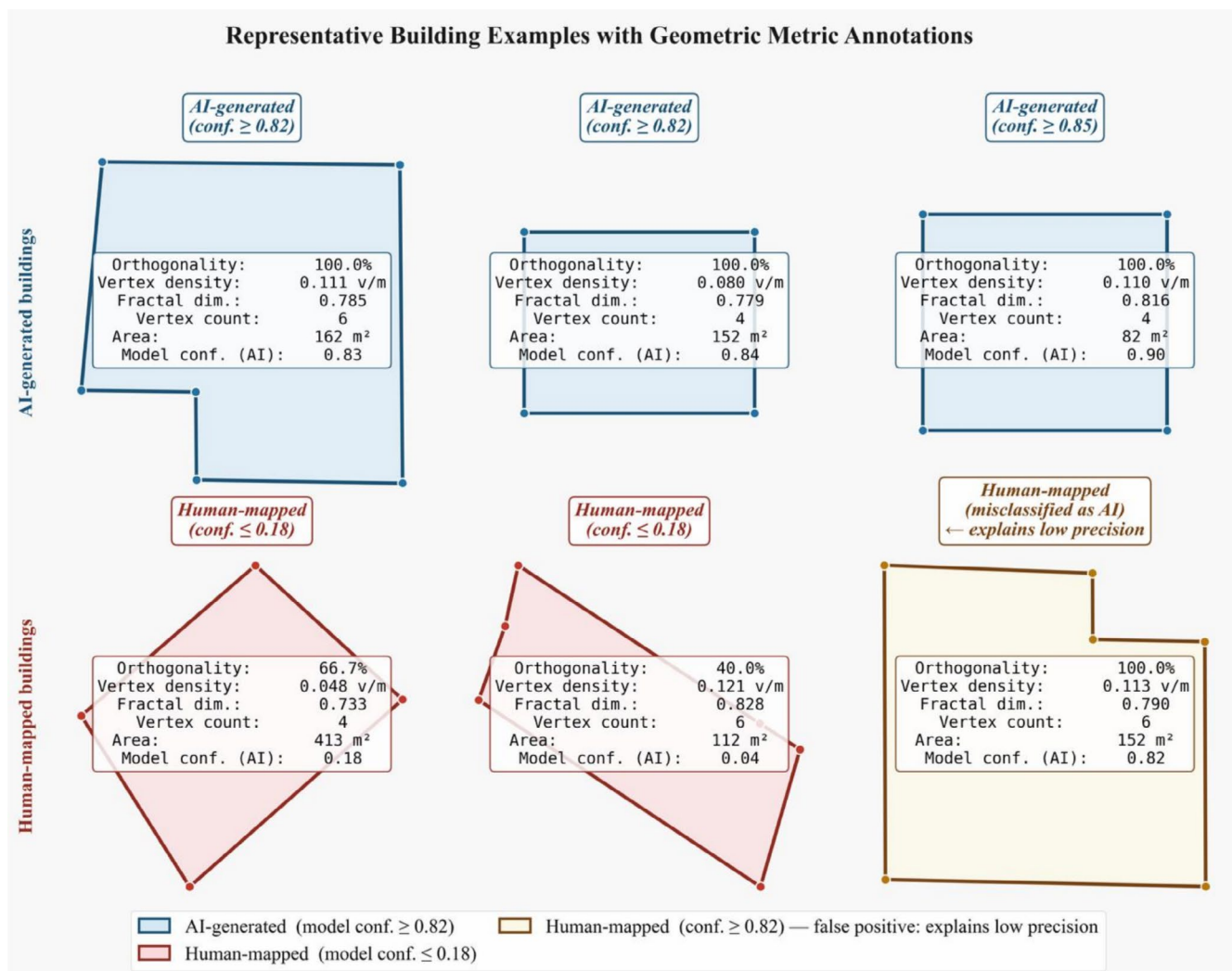


FIGURE 4 | Representative building footprints selected from the test set with geometric metric annotations. Row 1 shows AI-generated buildings classified with high model confidence (≥ 0.82); Row 2 shows human-mapped buildings—two correctly classified (confidence ≤ 0.18) and one misclassified as AI-generated (confidence = 0.82, bottom right). Vertex markers (●) indicate digitization granularity.

AI-generated and human-mapped buildings for 30 of the 32 geometric features examined (Figure 5). However, effect size analysis provides a more meaningful assessment of practical importance given the large sample size (Table 4). The analysis revealed that 8 features exhibited medium effect sizes ($0.3 < |r| \leq 0.5$), indicating substantial practical distinction between the two categories. An additional 11 features showed small effects ($0.1 < |r| \leq 0.3$), suggesting detectable but modest differences. The remaining 13 features showed negligible practical differences ($|r| \leq 0.1$) despite achieving statistical significance, a pattern expected given the large sample size of 244,974 AI-generated and 8,871,100 human-mapped buildings. These results demonstrate that morphometric differences between the two categories are both statistically significant and practically meaningful for a subset of features, while many features show minimal practical distinction despite statistical significance.

Positive effect sizes indicate AI buildings have lower values (vertex density, fractal dimension, perimeter-area ratio), reflecting simplified geometry, while negative values show AI buildings have higher values (area, vertex efficiency), indicating larger

buildings with economical vertex placement. Vertex-related features emerged as the strongest discriminators. Vertex density showed the largest effect size ($r = 0.366$, medium effect), with AI-generated buildings exhibiting significantly lower vertex density (median = 0.096 vertices/m) compared to human-digitized structures (median = 0.129 vertices/m), indicating that AI buildings use fewer vertices per unit perimeter length. Conversely, vertex efficiency (which measures area coverage per squared vertex count) was significantly higher in AI buildings (median = 5.84) than human buildings (median = 3.21, $r = -0.355$).

Boundary complexity measures revealed systematic differences in shape characteristics. The fractal dimension (standard method) was significantly lower in AI buildings (median = 0.785) compared with human buildings (median = 0.814, $r = 0.354$, medium effect). Similarly, the perimeter-area ratio was lower in AI buildings (median = 0.330) than human buildings (median = 0.419, $r = 0.347$). Size-related features showed consistent patterns, with AI-generated buildings being significantly larger than human-digitized structures. The median

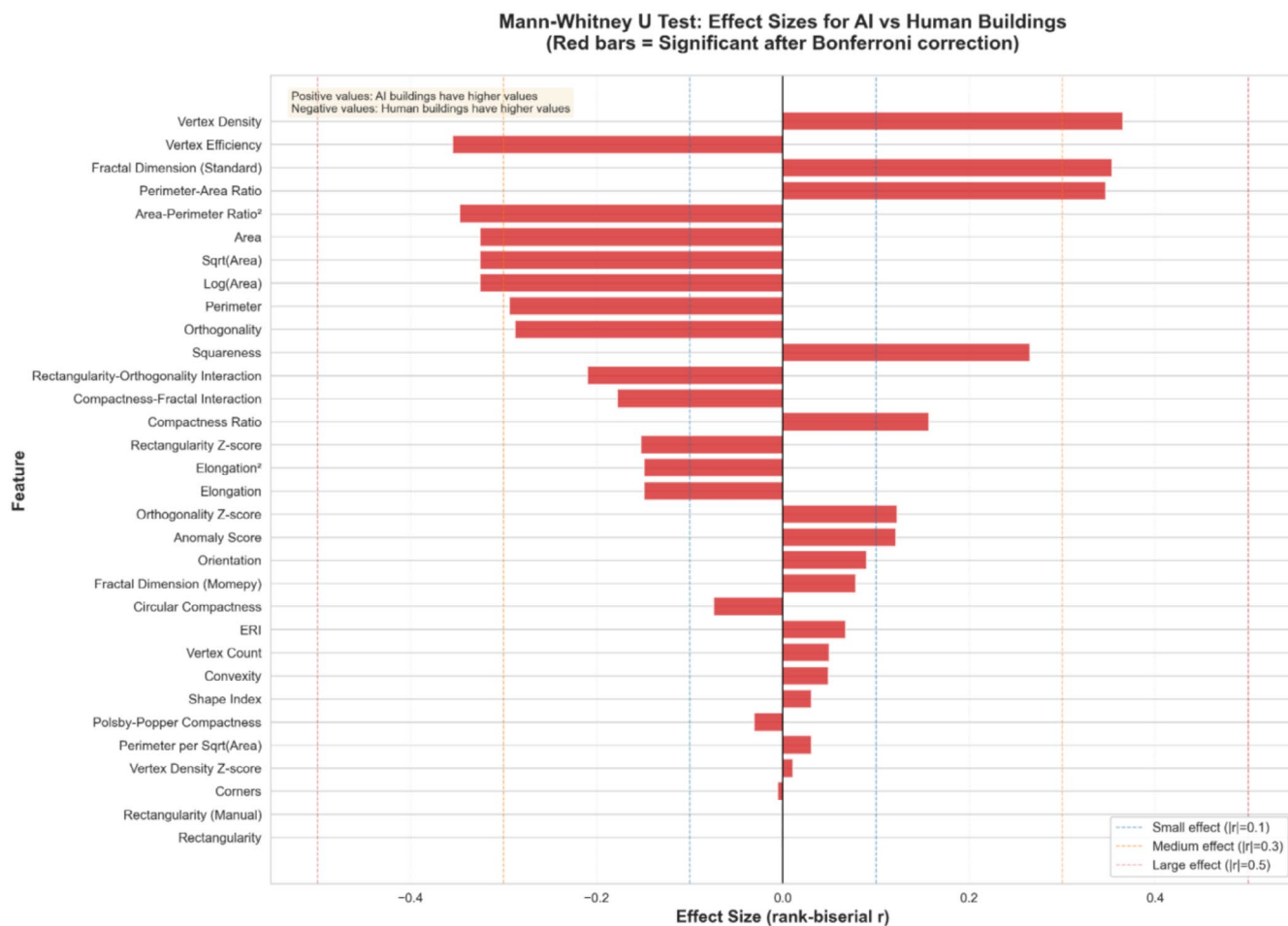


FIGURE 5 | Mann-Whitney U test effect sizes (rank-biserial r) for all 32 geometric features. Red bars indicate significance after Bonferroni correction ($p < 0.00156$). Given the large sample size, statistical significance is expected even for small differences; practical importance is better assessed through effect size magnitude, with only 8 features showing medium effects ($|r| > 0.3$).

area of AI buildings (171.35m^2) exceeded that of human buildings (107.30m^2 , $r = -0.325$, medium effect). This pattern held across all size transformations ($\text{Log}(\text{Area})$, $\text{Sqrt}(\text{Area})$). Orthogonality exhibited a statistically significant difference with a small effect size ($r = -0.288$), despite showing identical median values (1.000) for both groups due to the prevalence of perfectly rectangular buildings. Shape regularity features including squareness ($r = 0.266$), rectangularity-orthogonality interaction ($r = -0.210$), and compactness metrics demonstrated small but consistent effects. Notably, only two features (rectangularity and rectangularity_manual) failed to show significant differences after Bonferroni correction, likely due to their high correlation.

4.4 | Feature Importance and Geometric Discriminators

Feature importance analysis identified which geometric characteristics most strongly distinguish AI from human contributions (Figure 6). The top 15 most important features collectively accounted for approximately 85% of total model importance, with the remaining features contributing relatively minor discriminative power.

Orthogonality z-score emerged as the single most important feature, accounting for 18.16% of total importance in the LightGBM model and 5.90% in the Random Forest model. The raw orthogonality score ranked sixth overall (4.67% importance). Vertex-related metrics constituted critical complexity indicators: vertex density (5.00% LGBM importance, 13.62% RF importance), vertex count (6.24% LGBM, 2.17% RF), and vertex efficiency ranked fifth in the Random Forest model (11.38%). Orientation was the second-most important feature in the LightGBM model (15.91%). Shape regularity metrics (including squareness (7.74% average importance), rectangularity-orthogonality interaction (3.54%), and convexity (2.59%)) constituted additional discriminators between the two categories.

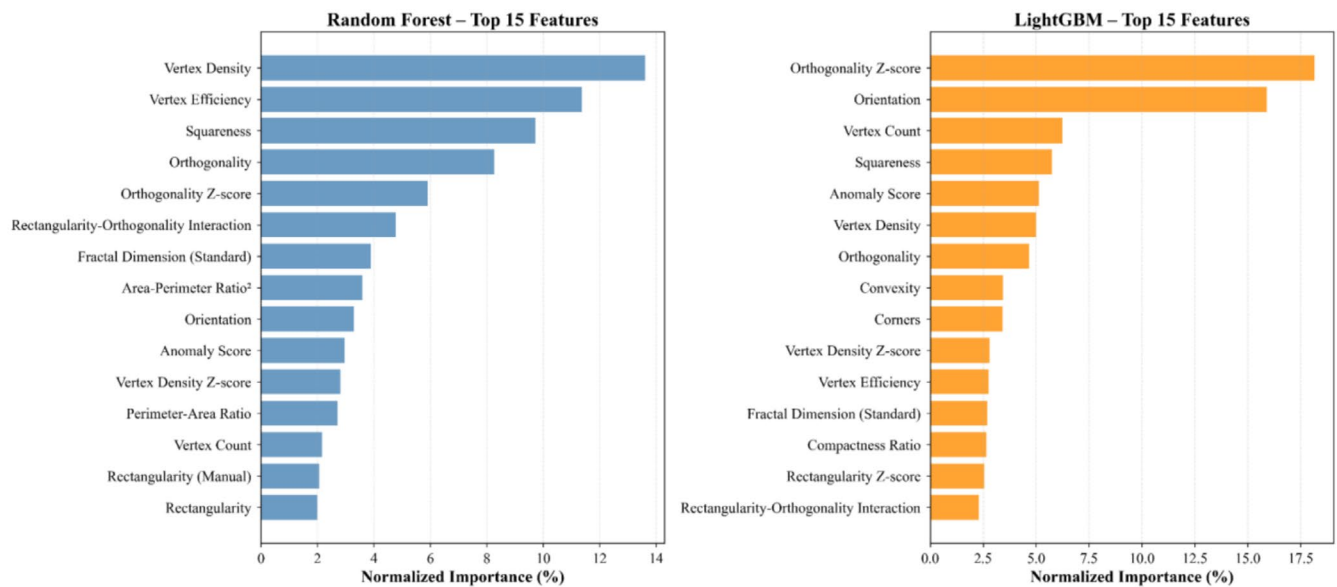
4.5 | Feature Category Analysis

To understand broader morphometric patterns, the 32 individual features were aggregated into four geometric categories based on their underlying morphological meaning as defined in Section 3.3 (Figure 7). Category-level importance was calculated by summing the normalized importance scores of constituent features within each group.

TABLE 4 | Top 10 geometric features ranked by effect size magnitude (rank-biserial r).

Feature	AI median	Human median	Effect size (r)	Interpretation	p
Vertex Density	0.0962	0.1290	0.366	Medium	< 0.001
Vertex Efficiency	5.8352	3.2137	−0.355	Medium	< 0.001
Fractal Dimension (standard)	0.7852	0.8140	0.354	Medium	< 0.001
Perimeter–Area Ratio	0.3300	0.4188	0.347	Medium	< 0.001
Area–Perimeter Ratio ²	9.1846	5.7016	−0.347	Medium	< 0.001
Area	171.35 m ²	107.30 m ²	−0.325	Medium	< 0.001
Sqrt(Area)	13.09	10.36	−0.325	Medium	< 0.001
Log(Area)	5.15	4.68	−0.325	Medium	< 0.001
Perimeter	55.99 m	44.82 m	−0.294	Small	< 0.001
Orthogonality	1.000	1.000	−0.288	Small	< 0.001

Note: Positive values: AI buildings higher; negative values: Human buildings higher. All features significant after Bonferroni correction ($p < 0.001$, $\alpha = 0.00156$). Effect size categories: Small (0.1–0.3), medium (0.3–0.5), large (> 0.5).

**FIGURE 6** | Top 15 features ranked by normalized importance (%) for Random Forest (blue) and LightGBM (orange) models.

Complexity and Angularity features dominated the classification task, accounting for 38.7% of total model importance. This category encompasses fractal dimensions, vertex metrics (vertex count, vertex density, vertex efficiency, corners), and elongation measurements that capture boundary intricacy and shape anisotropy. Derived Interactions and Anomaly Scores contributed 32.0% of total importance, making this the second most discriminative category. This grouping includes z-score normalized versions of key geometric properties (orthogonality, rectangularity, vertex density), interaction terms (rectangularity–orthogonality, compactness–fractal), and the composite anomaly score. Shape Regularity and Compactness features accounted for 20.2% of total importance, encompassing measures of rectangularity, orthogonality, squareness, convexity, and various compactness indices including Polsby–Popper compactness, circular compactness, and the Equivalent Rectangular Index. Size and Dimensions features contributed only 9.1% of total importance despite including fundamental properties

like area, perimeter, and their transformations, indicating that classification relies primarily on shape characteristics rather than scale.

4.6 | Geographic Performance Variation

Model performance varied substantially across cities and regions, revealing that classification effectiveness depends critically on local mapping contexts, urban morphologies, and AI adoption patterns.

4.6.1 | City-Level Performance

City-level analysis revealed balanced accuracy ranging from 57.50% (Berlin) to 80.60% (New York City), with a mean of 71.75% across the 15 cities (Figure 8). The best-performing cities

were predominantly from North America and Oceania, with New York City (80.60% balanced accuracy, 0.867 AUC), Sydney (79.90% balanced accuracy, 0.865 AUC), Cape Town (79.70% balanced accuracy, 0.864 AUC), and Toronto (79.10% balanced accuracy, 0.868 AUC) achieving the highest discrimination.

European cities exhibited the most variable performance, with London performing reasonably well (76.00% balanced accuracy, 0.831 AUC) while Berlin, Istanbul, and Budapest showed substantially lower discrimination (57.50%, 63.50%, and 70.20% balanced accuracy respectively). This heterogeneity may reflect differences in local digitization guidelines, mapper expertise levels, or the specific AI tools deployed in different European mapping campaigns.

Asian cities (Tokyo and Bangkok) demonstrated intermediate performance (59.40% and 71.60% balanced accuracy), though

sample sizes were smaller in these regions. The relatively lower performance in Tokyo (64.40% AUC) may reflect greater similarity between AI and human mapping practices in this highly organized urban environment, where both approaches produce geometrically regular building footprints. Tokyo exhibited particularly low recall (48%), meaning the model failed to identify more than half of AI-generated buildings in this city, despite achieving 59% balanced accuracy through high specificity on human buildings.

Precision showed even greater geographic variation, ranging from 33.10% (Berlin) to 55.00% (Cape Town), indicating that the proportion of false positives varied substantially by location. Cities with higher AI adoption rates (Seattle: 13.38%, Cairo: 15.70%, Sydney: 9.08%) tended to exhibit lower precision but higher recall, suggesting that models trained on imbalanced data prioritize sensitivity when AI buildings are more prevalent in the test set.

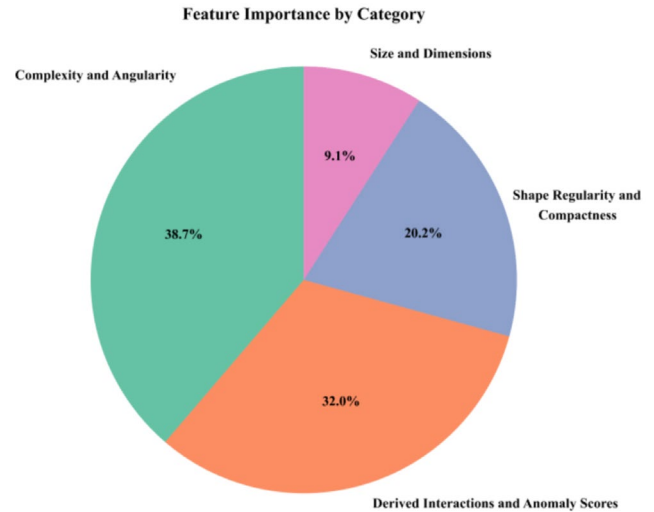


FIGURE 7 | Aggregated feature importance by morphometric category. Percentages represent summed importance scores of constituent features within each of the four geometric categories.

4.6.2 | Regional Performance Patterns

Regional aggregation revealed systematic geographic patterns in classification accuracy (Figure 9). Oceania achieved the highest balanced accuracy (79.90%), followed closely by Africa (76.90%), North America (75.20%), and South America (74.80%). Europe (70.00%) and Asia (67.50%) exhibited notably lower discrimination, suggesting that AI and human buildings in these regions share more similar geometric characteristics.

The regional AUC scores followed similar patterns, with Oceania (0.865), Africa (0.841), South America (0.835), and North America (0.823) all exceeding 0.82, while Europe (0.778) and Asia (0.736) showed reduced discriminative ability. These differences likely arise from multiple factors: (1) the specific AI tools deployed in different regions may produce more or less distinctive geometric signatures, (2) regional differences in urban morphology may create varying baseline levels of geometric regularity, and (3) local mapper cultures and digitization practices



FIGURE 8 | LightGBM performance metrics across 15 cities. Color intensity indicates score magnitude (darker = higher performance).

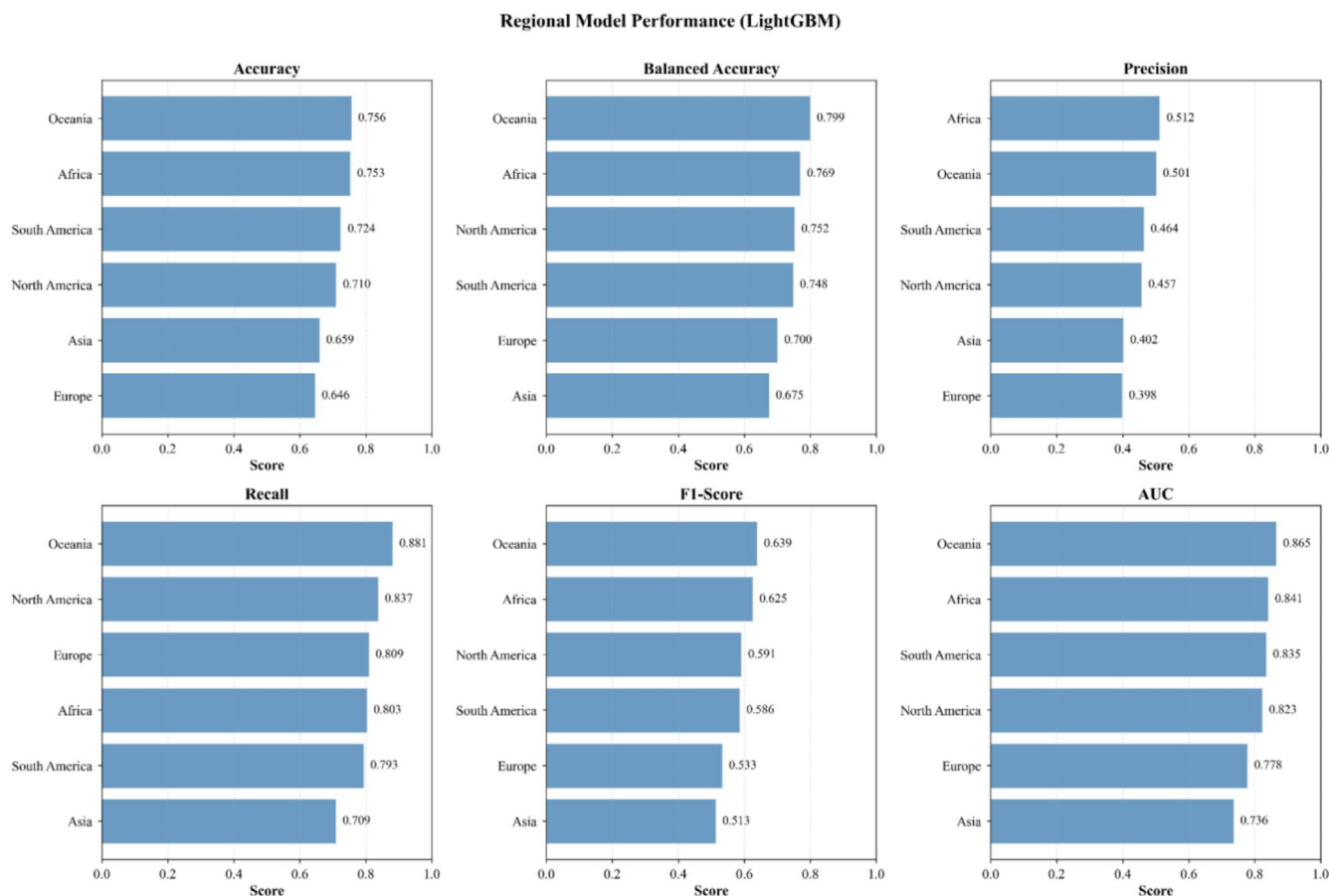


FIGURE 9 | Regional performance aggregation showing systematic geographic variation across six metrics.

may converge with or diverge from AI-generated patterns to different degrees.

The comparison of F1-scores across regions reveals that Oceania and Africa not only achieved higher median performance but also showed less variability, while Europe exhibited the widest performance range. Within-region heterogeneity was substantial, with European cities spanning 0.414 to 0.613 F1-score and North American cities ranging from 0.550 to 0.658, indicating that continental-scale patterns mask important city-specific factors. This heterogeneity within Europe suggests that classification difficulty varies substantially even within continental regions (Figure 10).

Geographic patterns in AI adoption and classification performance are visualized in Figure 11, which displays all 15 study cities with proportional symbols scaled by F1-score and colored by AI adoption rate using quantile classification. This visualization reveals several notable patterns. First, AI adoption rates vary dramatically across geographic contexts, with the highest concentrations in Cairo (15.70%) and Seattle (13.38%), while traditional OSM strongholds like Berlin and Tokyo show minimal AI presence (0.15% each). Second, classification performance does not correlate straightforwardly with AI adoption: Cape Town achieved the highest F1-score (0.660) with moderate AI adoption (2.50%), while Tokyo exhibited the lowest performance (0.413) despite having access to high-quality imagery and organized mapping communities. Third,

cities cluster regionally in both adoption patterns and performance, with North American and Oceanian cities generally achieving stronger discrimination than their European and Asian counterparts. These geographic disparities underscore that the effectiveness of morphometric classification depends not only on the presence of AI-generated buildings but also on local architectural styles, urban planning traditions, and the specific mapping tools and conventions employed in different regions.

5 | Discussion

The findings demonstrate that buildings labeled with AI source tags in OpenStreetMap exhibit distinctive geometric patterns detectable through machine learning, though substantial overlap between categories limits utility for definitive source attribution. Three insights emerge: algorithmic extraction produces systematic regularization in orthogonality and vertex distributions (Section 5.1), classification effectiveness varies substantially across geographic contexts (Section 5.2), and approximately one-third of human-mapped buildings exhibit AI-like signatures (Section 5.3). These patterns suggest that geometric morphometry can serve as a complementary approach to metadata-based quality assessment, though under realistic deployment conditions with 2.69% AI prevalence, positive predictive value constraints necessitate human verification rather than automated decision-making.

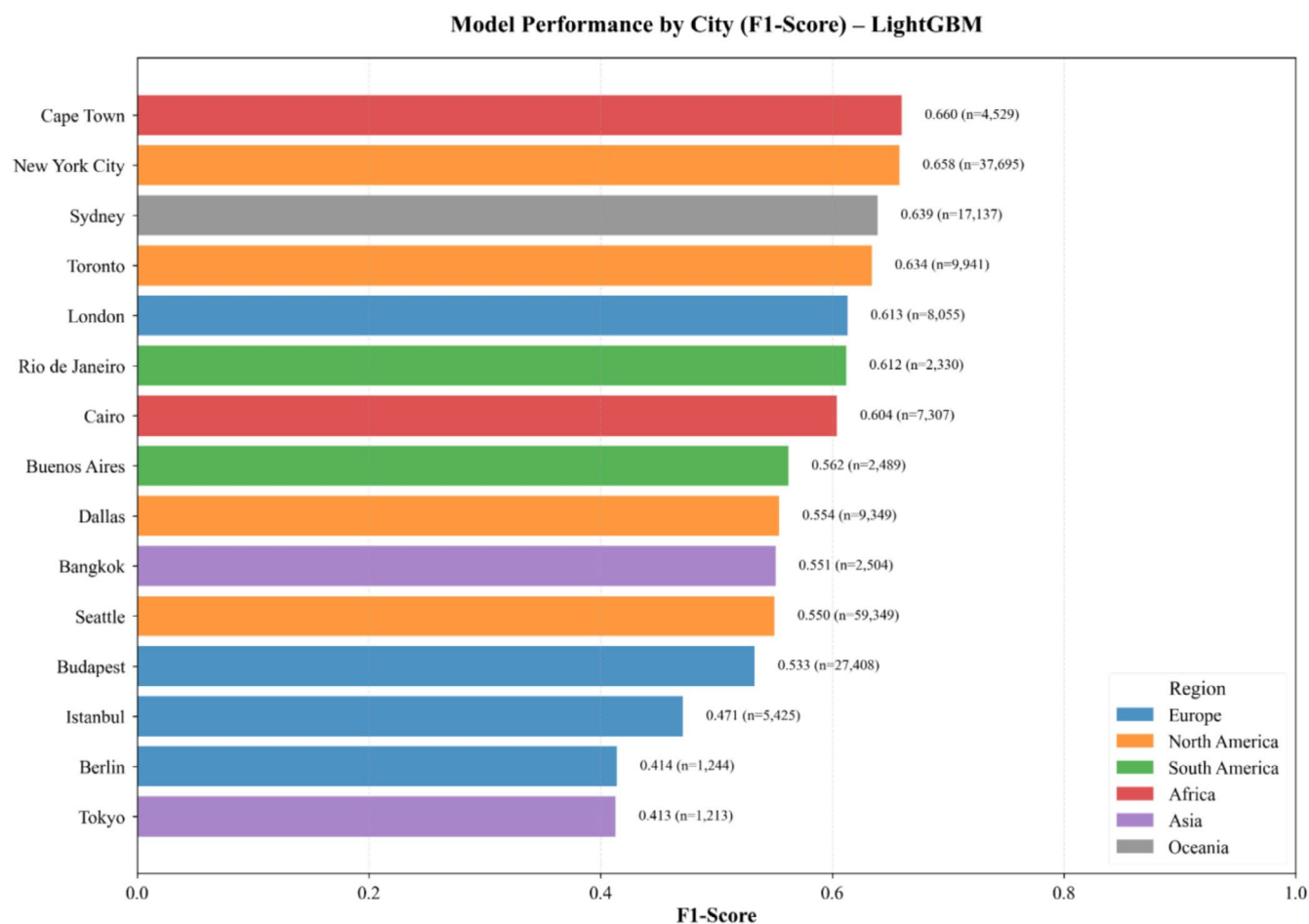


FIGURE 10 | City-level F1-scores with regional color-coding. Sample sizes in parentheses. Cities ordered by descending F1-score.

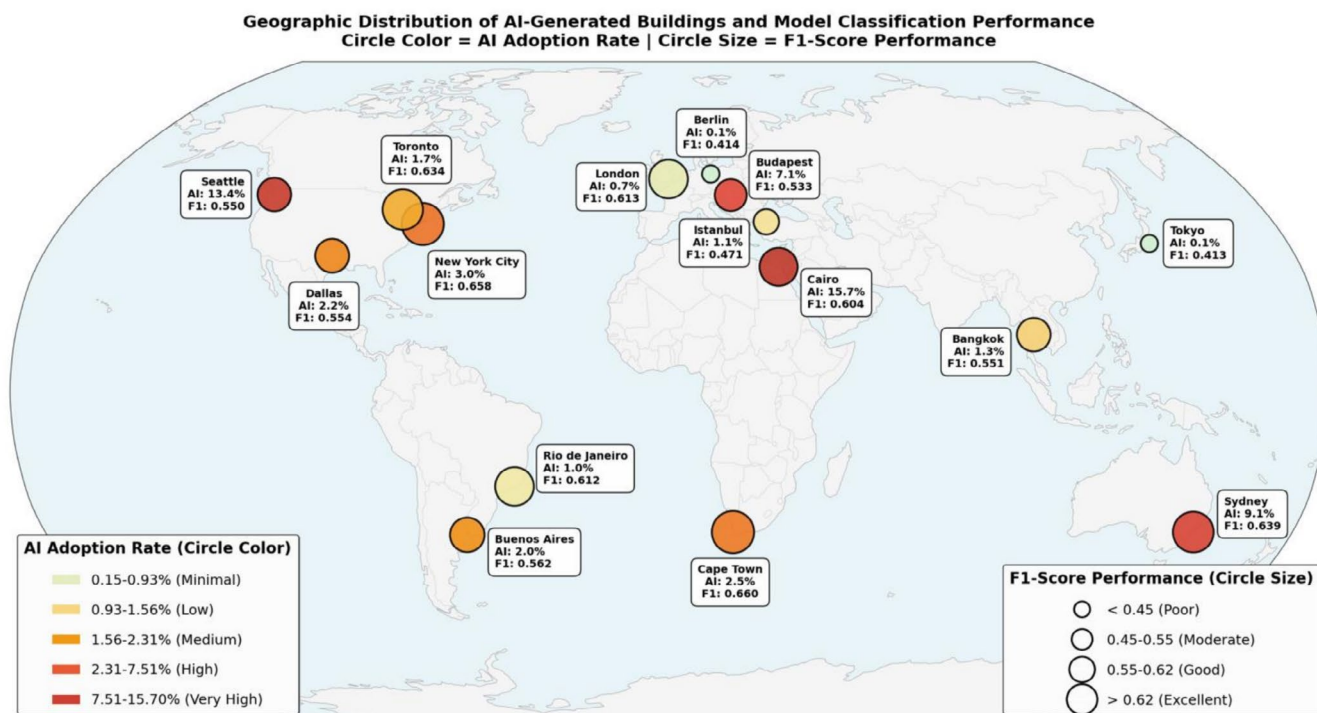


FIGURE 11 | Geographic distribution of AI-generated buildings and model classification performance across 15 study cities. Circle size represents F1-score performance (larger circles indicate better classification accuracy), while circle color indicates AI adoption rate (light green to dark red gradient). Cities are labeled with their AI adoption percentage and F1-score.

5.1 | Geometric Signatures and Algorithmic Regularization

The prominence of orthogonality measurements and vertex patterns as primary discriminators reveals how automated extraction systems leave distinctive geometric fingerprints in crowdsourced geodata. These patterns likely reflect the geometric regularization and shape simplification algorithms employed in computer vision models trained on satellite imagery, producing systematically different angular distributions and vertex spacing compared to human-digitized structures, even when both categories contain predominantly rectangular buildings.

These morphometric signatures transcend simple shape preferences and reveal fundamental differences in how algorithms versus humans represent spatial uncertainty and geometric detail. Human mappers introduce vertex placement variability that may capture subtle architectural details, compensate for imagery distortions, or simply reflect the natural inconsistency of manual digitization. In contrast, AI algorithms generate more evenly spaced vertices along building perimeters, creating geometrically regular representations that optimize for edge detection accuracy and computational efficiency rather than architectural fidelity. The observed lower fractal dimensions and reduced boundary complexity further confirm that automated extraction produces smoothed boundaries that prioritize regularity over detail preservation. It should also be noted that this signature represents an average across heterogeneous systems (Microsoft, Google, fAIr, and Meta/Maxar), each with distinct architectures and post-processing pipelines that may produce tool-specific patterns obscured by their aggregation (discussed in Section 3.7). Notably, AI-generated buildings are statistically larger than human-digitized structures (median 171 m² vs. 107 m²), the mechanism underlying this difference is unclear. However, since size-based features provide limited discriminative power for classification, the primary AI signature is encoded in shape regularization characteristics rather than scale.

The category-level importance hierarchy reveals fundamental differences in how geometric information distinguishes digitization methods. Complexity and angularity features, combined with derived interactions and anomaly scores, collectively account for the majority of discriminative power, while size and dimensions contribute minimally. The dominance of vertex-based metrics within the complexity category further indicates that digitization granularity (the density, spacing, and efficiency of vertex placement) encodes more information about algorithmic versus manual origin than traditional shape indices like compactness or rectangularity alone. This pattern suggests that future quality control systems should prioritize boundary complexity analysis when assessing data provenance. These morphometric patterns align with recent observations that different automated extraction architectures apply distinct geometric regularization strategies, producing systematically different boundary characteristics and vertex placement patterns (Niroshan and Carswell 2025a). As AI-assisted mapping workflows proliferate across OSM (Sun et al. 2026), understanding these architecture-specific geometric signatures becomes increasingly important for quality assessment.

The importance of derived interactions and anomaly scores among the top discriminative features reveals that AI-generated buildings are distinguishable not merely by their absolute geometric properties but by how they deviate from global statistical patterns. The composite anomaly score, which aggregates z-standardized deviations across multiple geometric dimensions, captures buildings that simultaneously exhibit extreme orthogonality, regularity, and vertex efficiency: a multidimensional signature more diagnostic than any single metric. This finding suggests that AI algorithms produce buildings occupying distinctive regions of morphometric feature space characterized by correlated extremes across multiple shape descriptors. The prevalence of this multi-dimensional signature in machine learning feature importance reflects the fact that algorithmic extraction leaves distinctive fingerprints across multiple geometric dimensions simultaneously, rather than through any single morphometric property.

5.2 | Geographic Heterogeneity and Contextual Factors

The substantial geographic variation in both AI adoption and model performance reveals that the integration of automated tools into OpenStreetMap reflects strategic deployment rather than uniform global application. The concentration of AI contributions in specific regions (particularly Cairo and Seattle compared to minimal presence in Berlin and Tokyo) likely reflects targeted humanitarian mapping campaigns and organized efforts to improve base map coverage in data-sparse regions. These patterns raise fundamental questions about the future trajectory of crowdsourced mapping: whether AI tools will democratize cartography by reducing barriers to entry or create new forms of geographic inequality by concentrating automated resources in specific contexts while others remain dependent on volunteer effort.

Classification effectiveness depends critically on local mapping contexts and urban morphologies, revealing that geometric signatures are fundamentally contextual rather than universal. Several mechanisms likely underlie this geographic heterogeneity. Regional differences in architectural styles and urban planning traditions establish varying baseline levels of geometric regularity: cities with more uniformly rectangular building stock create less distinction between algorithmic and human-produced forms. Local mapping conventions and community guidelines shape how human contributors digitize buildings, fostering region-specific “styles” that either converge with or diverge from AI-generated patterns. Different AI tools deployed across regions may employ distinct regularization approaches, producing varying geometric signatures. These mechanisms suggest that what constitutes an “AI-like” geometric signature is not purely algorithmic but emerges from the interaction between local building morphologies, established mapping practices, and tool-specific characteristics. Notably, geographic performance variation occurs despite uniform model architecture and hyperparameters across all regions, indicating that classification difficulty is driven by inherent data characteristics rather than parameter configuration choices.

This geographic contingency carries significant implications for practical deployment. Cities with minimal AI adoption present a paradox: the handful of AI-generated buildings in these contexts are exceptionally difficult to distinguish, likely because highly organized urban environments with uniform architectural conventions make both human and algorithmic approaches converge toward similar geometric solutions. Conversely, stronger performance in regions with higher AI adoption may reflect either that algorithmic patterns become more detectable when present at scale, or that these urban contexts feature greater morphological diversity that provides clearer contrast between automated and manual digitization. This heterogeneity reveals a fundamental limitation: a detection system trained on global data may exhibit systematic bias in its effectiveness across regions, potentially creating disparities in how AI contributions are monitored across different mapping communities. These disparities reflect broader patterns in how GeoAI innovations diffuse unevenly across mapping communities, often reinforcing rather than ameliorating existing geographic inequalities (Sun et al. 2026). Practical implementation therefore requires acknowledging that no single classification threshold can be universally appropriate, and that deploying geometric analysis in quality control workflows demands region-specific calibration and transparent communication about where the approach succeeds and where it fails.

5.3 | Methodological Convergence and the Future of Human-AI Collaboration

The approximate one-third of human-mapped buildings exhibiting AI-like geometric signatures represents more than a classification challenge: It reveals a fundamental transformation in how mapping is practiced. This substantial overlap challenges the assumption that “human-mapped” and “AI-generated” represent distinct methodological categories, suggesting instead that contemporary mapping operates along a continuum where geometric regularization has become an implicit standard regardless of origin.

The geometric convergence reflects an important shift in mapping norms and practices. The proliferation of editing guidelines, validation tools, and quality standards across OpenStreetMap has established geometric regularity as an expected output, not an algorithmic byproduct. When experienced mappers intentionally regularize building geometries for cartographic clarity, or when novice contributors receive peer feedback encouraging cleaner outlines, both are responding to the same normative pressure: the belief that systematized, regular geometries represent better data. Algorithmic systems optimize for this same criterion through computational constraints. This shared destination, geometric regularity, emerges not from human imitation of algorithms or vice versa, but from convergent solutions to the same problem: representing spatial features from imperfect imagery. The implication is profound: geometric regularity is no longer a signature of machine origin, but rather a quality standard that human expertise and algorithmic processing now achieve independently.

Yet disentangling genuine methodological convergence from epistemological artifacts proves impossible without independent ground truth. OSM’s design creates fundamental ambiguities:

the version history system captures only the most recent editor, meaning AI-generated buildings modified by humans appear as “human” contributions, and human-digitized buildings subjected to algorithmic refinement tools acquire machine-like signatures through secondary processing. Individual variation in human digitization styles further complicates interpretation: mappers with naturally systematic approaches produce regular geometries not through external influence but through inherent practice variation. These structural limitations mean morphometric analysis reveals what source tags claim rather than what actually occurred. The observed overlap may reflect genuine convergence, label uncertainty from hybrid workflows, incompleteness in version tracking, or some inseparable combination: Questions that cannot be resolved through geometric features alone.

This epistemological constraint fundamentally reframes the research question. Rather than asking “which method produces better data?” or “can we definitively distinguish human from AI contributions?” (questions that geometric analysis cannot answer) the productive inquiry becomes: “how do human and algorithmic mapping systems productively coexist, and which approaches serve different contexts and purposes?” This reframing acknowledges what the data reveals: that geometric regularity is increasingly a shared standard, not a discriminator. AI tools may prove most valuable for rapid base map completion in data-sparse regions where comprehensive geometric coverage outweighs architectural nuance. Human mappers remain irreplaceable for complex urban environments, historically significant structures, and applications requiring geometric fidelity grounded in local knowledge and contextual judgment. The challenge for crowdsourced mapping communities is not determining which method to privilege, but developing governance frameworks, transparency standards, and contributor guidelines that support productive human-AI collaboration where automation enhances rather than displaces community-driven cartography. This requires honest acknowledgment that the boundary between human and AI contributions is increasingly blurred, and that future data quality depends less on maintaining categorical purity than on understanding what different approaches contribute and maintaining transparency about provenance, limitations, and appropriate use cases.

5.4 | Implications for OSM Quality Control and Governance

Geometric classification addresses three operational needs in OSM quality assurance: prioritizing validation effort toward geometrically anomalous buildings, quantifying regional heterogeneity in mapping methodologies, and providing independent provenance assessment when metadata is incomplete. However, building on the epistemological constraints established in Section 5.3, geometric classification’s practical utility depends critically on understanding the precision-recall tradeoff under realistic deployment conditions. The high recall (82.9%) achievable on balanced test sets becomes fundamentally constrained by low precision when applied to real-world data. Under realistic AI prevalence (2.69%), positive predictive value drops to approximately 6%: meaning roughly 94% of buildings flagged as potentially AI-generated would be false positives. This asymmetry is

not a limitation to overcome through improved models, but an inherent property of the classification problem when base rates are so skewed. Automated geometric analysis can never serve as a reliable filter for provenance attribution; it can only generate candidates for human review.

Recognizing this constraint reframes geometric classification's appropriate role. Rather than answering "is this building AI-generated?," it can reliably answer "does this building exhibit geometric patterns associated with algorithmic extraction?" The distinction matters operationally. Geometric analysis becomes valuable for exploratory and prioritization workflows: flagging buildings for human verification, identifying regions where geometric heterogeneity suggests mixed sources, tracking temporal shifts in mapping methodologies, and revealing where algorithmic signatures are concentrated. These applications serve as complementary inputs to broader quality assessment frameworks integrating changeset metadata, contributor histories, and domain expertise: not as standalone decision systems. For downstream OSM users (including urban planners, disaster responders, and researchers), these findings suggest that geographic regions exhibiting higher geometric heterogeneity warrant additional scrutiny when data provenance affects analytical outcomes. Notably, this study does not address whether AI-generated buildings are geometrically more accurate than human-mapped ones; that question requires comparison against authoritative ground truth datasets and falls outside this intrinsic morphometric framework.

Practical implementation requires transparent acknowledgment of what geometric analysis can and cannot accomplish. It can prioritize attention toward buildings exhibiting unusual geometric combinations, but cannot determine provenance definitively. It can reveal geographic patterns in geometric regularity, but cannot prove those patterns resulted from specific tools. It can support comparative studies of how different source categories diverge geometrically, but cannot validate source tag accuracy. Communities deploying geometric analysis must understand these boundaries and communicate them clearly to mappers and quality control teams. The approach's value lies not in automating provenance decisions, but in enabling human judgment informed by geometric evidence: treating pattern detection as one signal among many within quality control workflows rather than as definitive proof of origin.

The challenge of maintaining transparency extends beyond geometric analysis. As AI-generated and human-verified contributions become increasingly difficult to distinguish in crowdsourced databases, quality control systems must address not only geometric accuracy but also provenance documentation and metadata transparency (Andorful et al. 2025; Niroshan and Carswell 2025b). This underscores that geometric classification serves as one component within broader quality assurance frameworks rather than a standalone solution.

5.5 | Future Directions and the Limits of Morphometric Analysis

This research has revealed the potential and the boundaries of geometric analysis for understanding mapping provenance.

Future directions should prioritize questions emerging directly from these constraints rather than pursuing incremental extensions.

Ground truth validation emerges as a critical prerequisite for validating whether morphometric signatures meaningfully correspond to actual provenance. This study assumes that OSM source tags accurately capture provenance, but this assumption has never been independently tested. Acquiring authoritative provenance data (through controlled studies where building creation and editing are fully documented, interviews with mapping communities, or analysis of projects with explicit metadata) would determine whether morphometric signatures actually correspond to digitization method or primarily reflect quality standards transcending source categories. This validation is essential before investing heavily in classification systems, since the entire endeavor rests on the premise that geometric patterns encode meaningful information about origin.

Secondary to validation, tool-specific analysis could advance understanding of algorithmic signatures. If metadata becomes available about which AI systems were deployed in which regions, investigating whether different tools produce distinctive geometric patterns could reveal whether the "AI signature" is truly universal or specific to particular algorithms and training regimes. Similarly, temporal analysis tracking geometric evolution as buildings are edited over time could illuminate whether apparent convergence reflects genuine standardization of practice or artifacts of labeling ambiguity in hybrid workflows. These investigations could distinguish methodological convergence from epistemological limitation.

Expanding analysis to underrepresented regions and other feature types (roads, waterways, land use boundaries) would test whether findings generalize beyond urban buildings in well-mapped areas. Investigating whether different automated extraction architectures produce distinct geometric signatures could reveal whether the patterns identified here represent universal AI characteristics or tool-specific artifacts. However, such expansions should prioritize explicit hypothesis-testing about contextual factors shaping geometric patterns rather than being pursued simply for geographic coverage.

These priorities reflect a fundamental insight: the question "can we detect AI?" has been sufficiently answered. The more productive question now becomes "what do geometric patterns reveal about how mapping is actually practiced, and how can we design governance systems that acknowledge this reality rather than seeking to eliminate or definitively categorize it?" Future research should shift from detection toward understanding: from asking "which buildings are AI?" to asking "how are human and algorithmic practices entangled in crowdsourced data, and what does this mean for quality, transparency, and community agency in the age of automated mapping?"

6 | Conclusions

This research reveals both the detectability and the fundamental insufficiency of geometric analysis for determining building provenance in OpenStreetMap. Buildings labeled with AI

source tags exhibit statistically distinct morphometric signatures (higher orthogonality, reduced vertex density, and simplified boundary complexity) that can be computationally identified. Yet approximately one-third of human-mapped buildings exhibit these same patterns, a finding that challenges binary categorization and exposes a deeper truth: Geometric regularization has become a shared quality standard achievable through both algorithmic constraint and human expertise. Moreover, this detectability varies substantially across geographic contexts, with classification accuracy ranging from 67.5% in Asia to 79.9% in Oceania, revealing that what constitutes an “AI signature” depends critically on local mapping conventions, architectural styles, and tool deployment patterns. These findings collectively demonstrate that contemporary crowdsourced mapping does not separate neatly into “human” and “AI” categories, but operates along a continuum where human judgment and machine intelligence are increasingly entangled.

This research establishes a methodological foundation for detecting geometric patterns associated with algorithmic extraction. However, this capability comes with a critical epistemological boundary: morphometric analysis reveals correlations with source tags rather than definitive provenance. OSM’s version history tracks only the most recent editor, hybrid workflows remain undocumented, and individual variation in human practice creates natural convergence with algorithmic outputs. These structural limitations mean that geometric classification cannot answer “which buildings are AI?” but only “which buildings exhibit AI-associated patterns?” This distinction is essential. The approach’s value lies not in automated provenance attribution but in supporting human judgment: flagging candidates for review, identifying regional heterogeneity, and revealing how different mapping approaches coexist within shared geographic spaces.

As artificial intelligence becomes embedded in crowdsourced cartography, understanding this hybrid landscape becomes essential to ensuring AI tools enhance rather than compromise the community-driven ethos that has made OpenStreetMap transformative. The challenge is not maintaining categorical purity between “human” and “AI” contributions, but developing governance frameworks that acknowledge ambiguity while maintaining transparency about data provenance, tool usage, and appropriate use cases. This requires honest recognition that the boundary between human and algorithmic mapping is increasingly blurred: not as a failure of classification systems, but as a reflection of how contemporary mapping is actually practiced. Future research should prioritize ground truth validation to determine whether morphometric signatures correspond to actual digitization provenance or primarily reflect quality standards transcending source categories, coupled with deeper investigation of how human and algorithmic practices productively coexist in reshaping crowdsourced geographic information in the age of automation.

Acknowledgments

The author has nothing to report.

Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

The code supporting this research is openly available at <https://github.com/carto-pi/AI-vs-Human-Buildings-in-OSM>. The building footprint data used in this study are freely available from OpenStreetMap (<https://www.openstreetmap.org/>) under the Open Database License (ODbL).

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